

Link State Routing

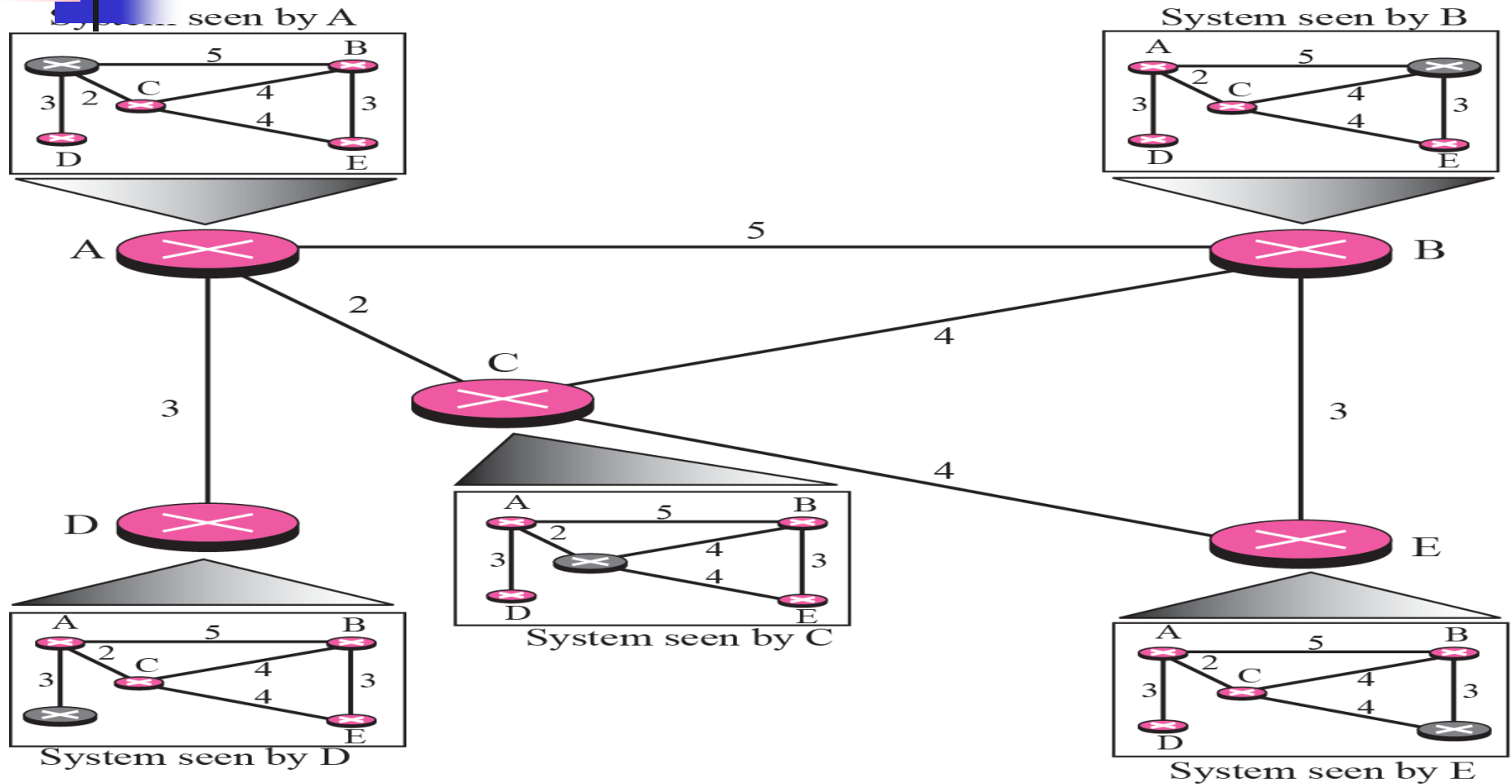
LINK STATE ROUTING

In link state routing, each node in the domain has the entire topology of the domain—

- the list of nodes and links,
- how they are connected including the type, cost (metric)
- the condition of the links (up or down)

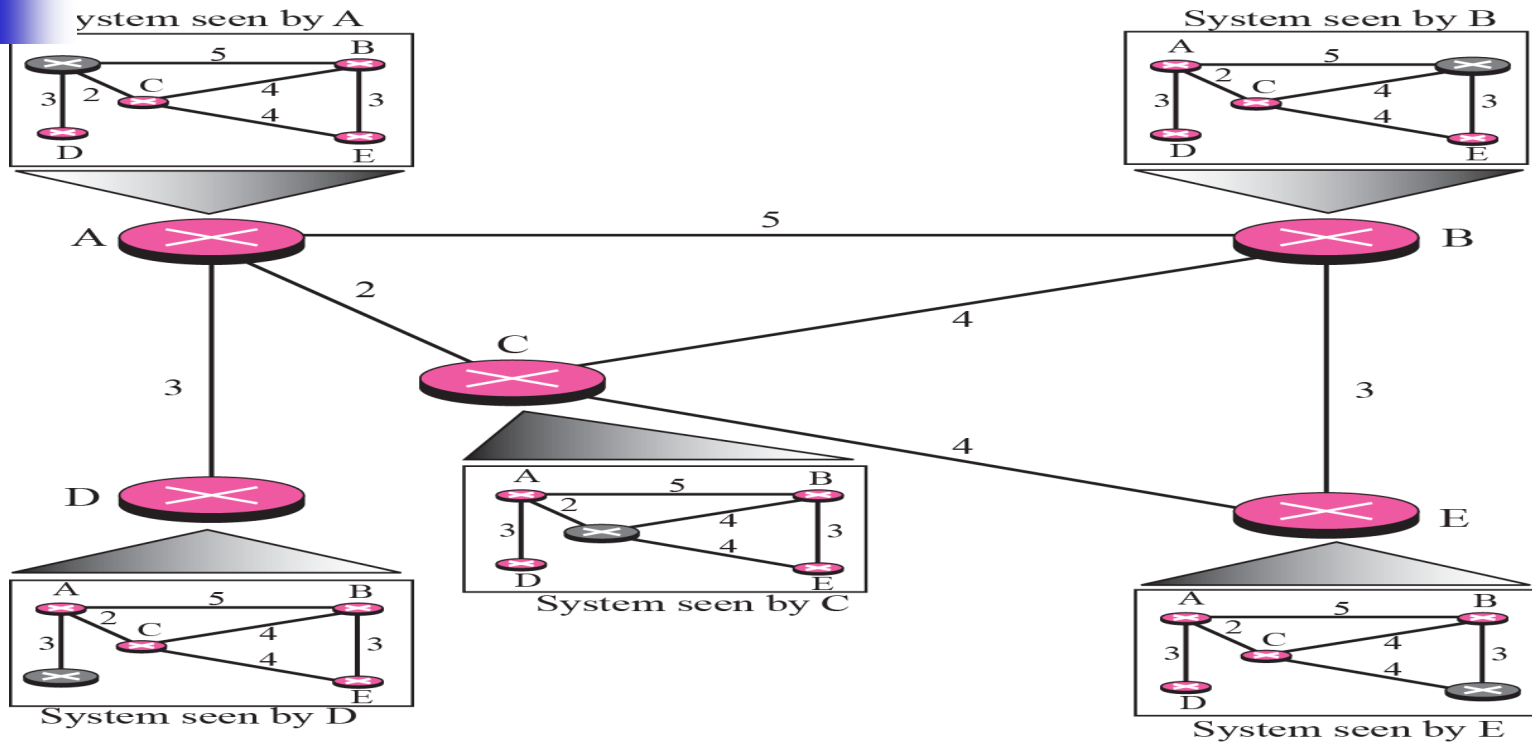
Node use the Dijkstra algorithm to build a routing table.

Concept of Link state routing



Each router in an LSR network maintains a complete, synchronized view of the network topology. This includes information about all links, their associated costs, and the characteristics of each link (like bandwidth, reliability, or delay).

Concept of Link state routing

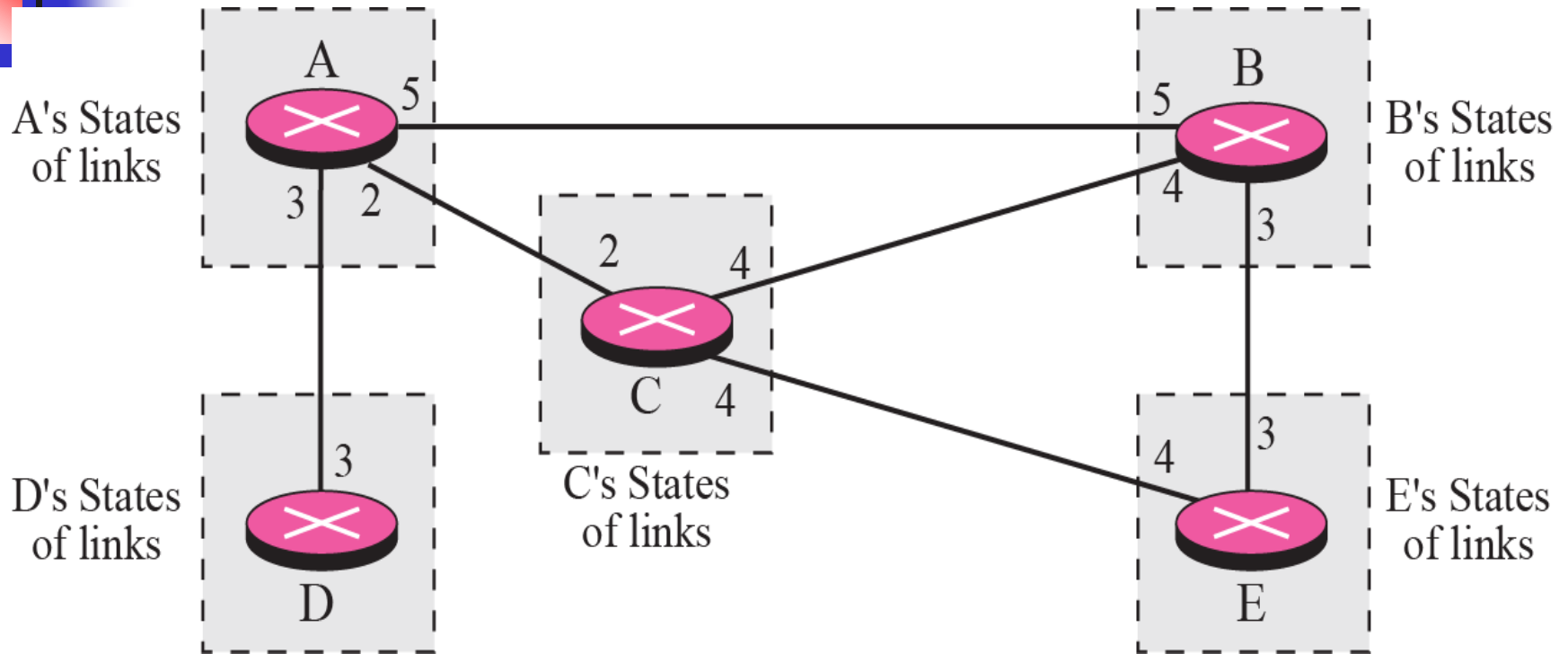


Routing protocols use the 'cost' of a link as a metric. The shortest path is then the path with the least penalty or least cost.

Typically, costs are automatically assigned based on the bandwidth of a link

- cost can be based on a variety of factors such as security levels, traffic or the state of the link.
- Other factors may be Path reliability, load, speed, latency, packet loss

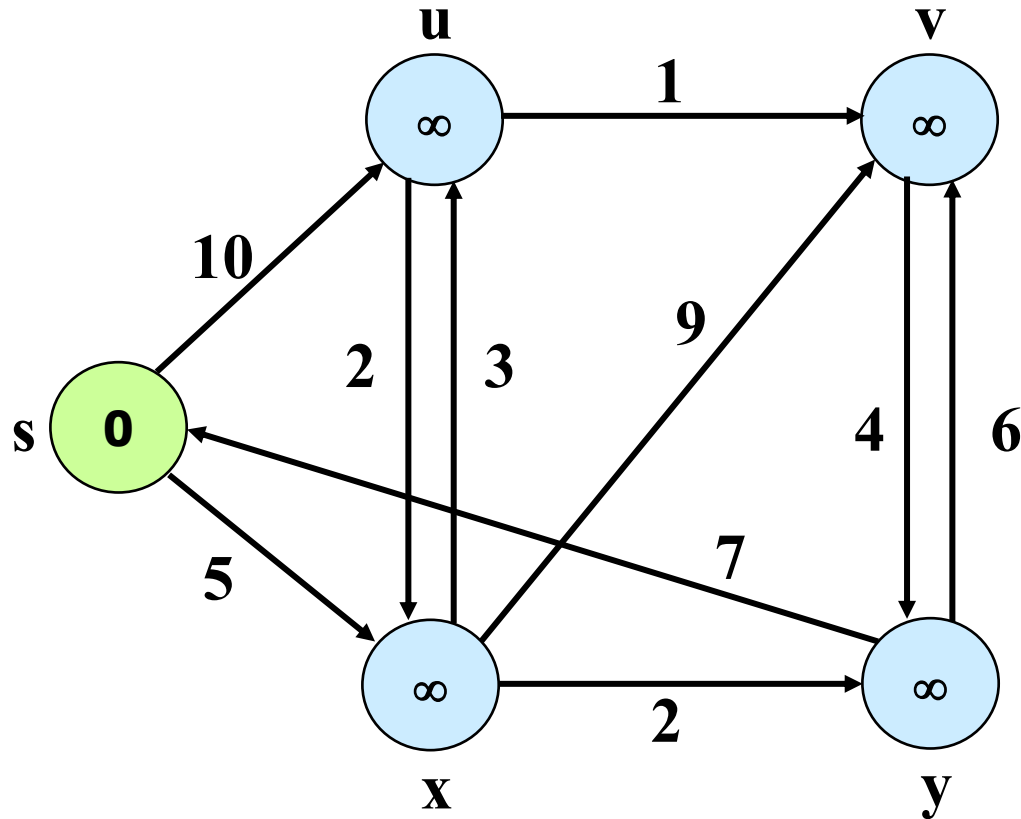
Link state knowledge



The metric is an arbitrary number assigned to the link, most importantly describing its bandwidth, that causes traffic to follow high bandwidth paths.

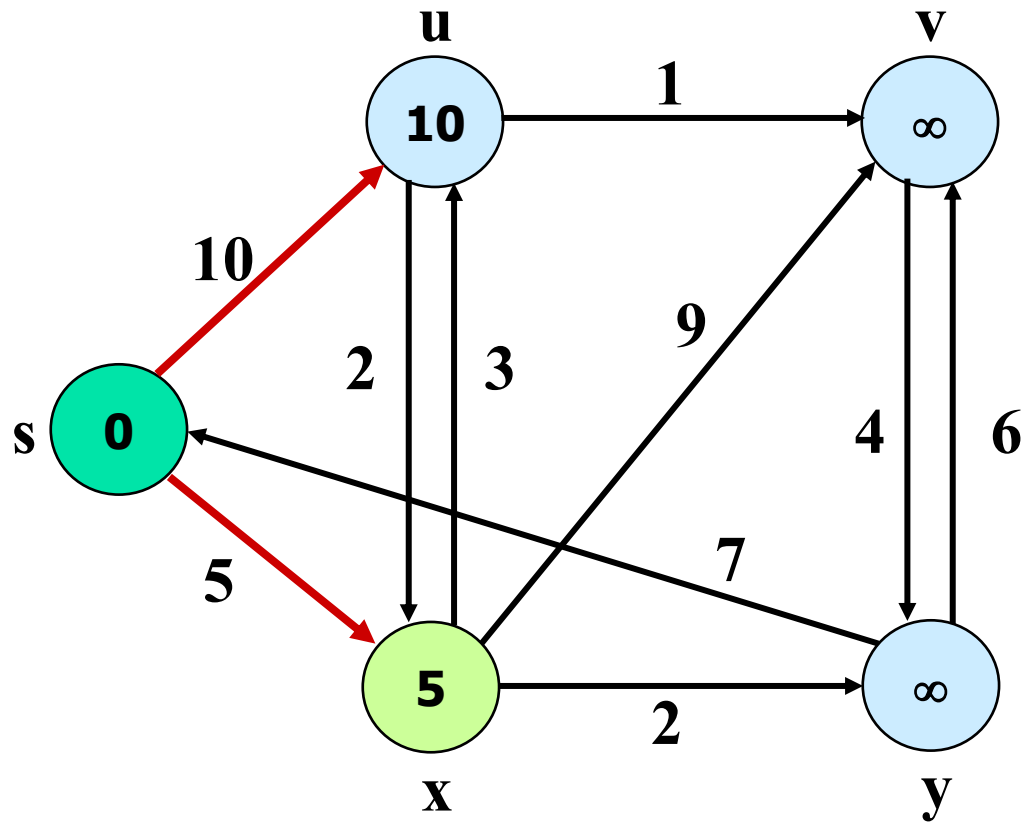
- **Bandwidth:** Higher bandwidth links are generally considered to have a lower cost because they can handle more traffic with less delay.
- **Delay:** Links with higher latency (longer processing times) or longer propagation delays might be assigned a higher cost.
- **Reliability:** More reliable links could have a lower cost.
- **Load:** Highly loaded links might be assigned a higher cost to discourage traffic from using them.
- **Other factors:** Network administrators can also assign arbitrary costs based on their specific needs and policies.

Example- Dijkstra Method



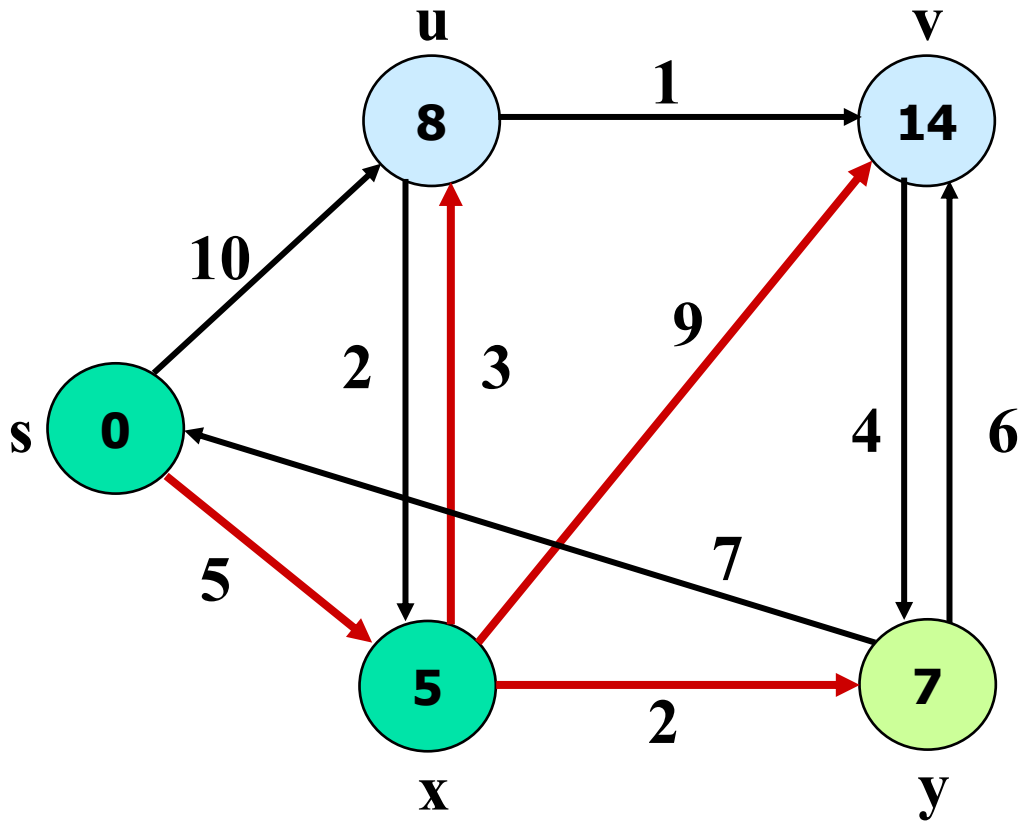
s	u	v	x	y
0	∞	∞	∞	∞

Example



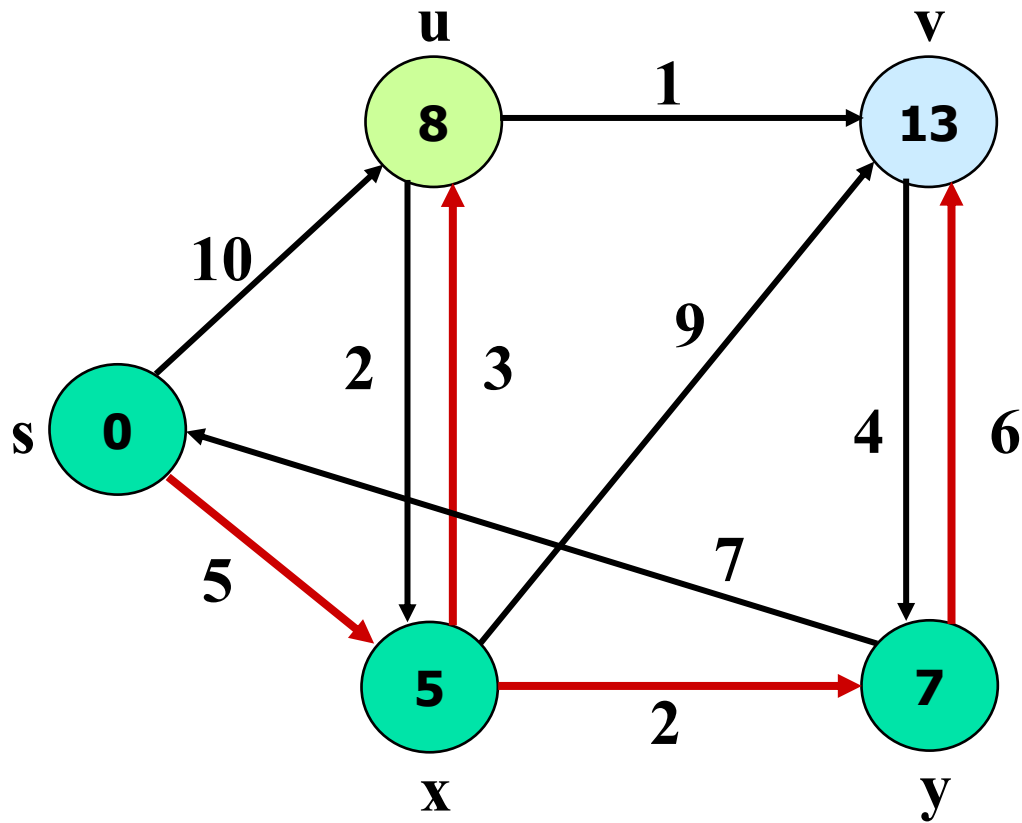
s	u	v	x	y
0	∞	∞	∞	∞
-	10s	-	5s	-

Example



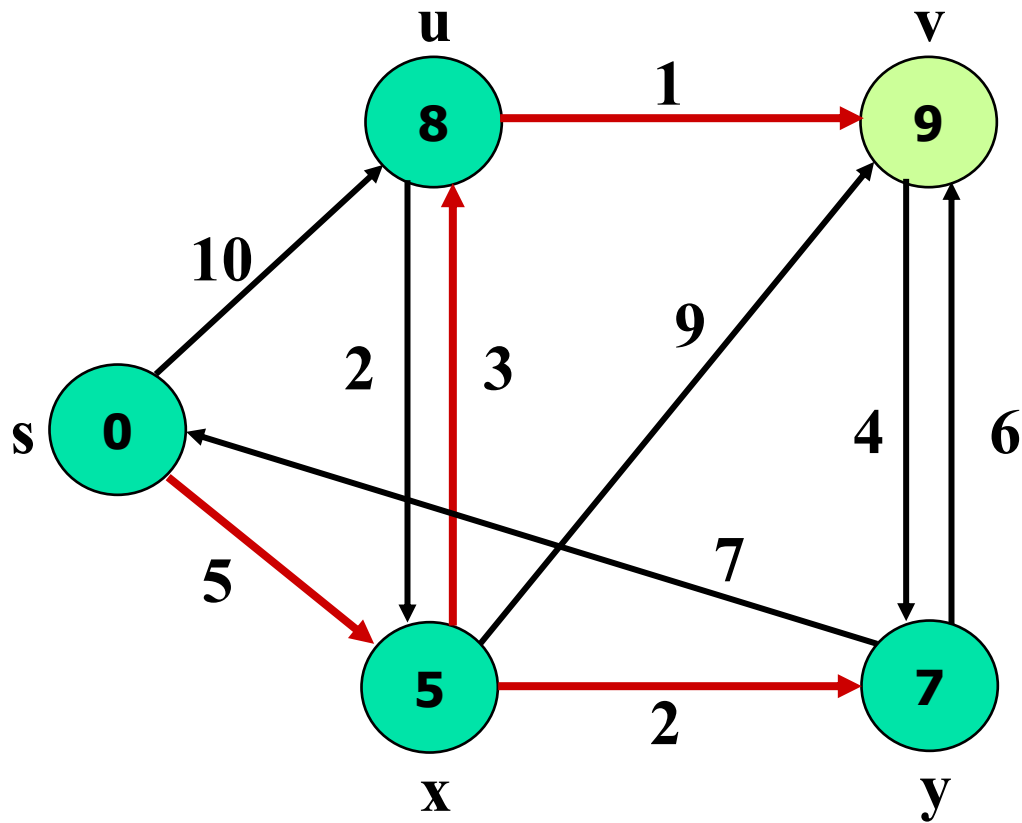
s	u	v	x	y
0	∞	∞	∞	∞
-	10 s	-	5s	-
-	8x	14 x		7x

Example



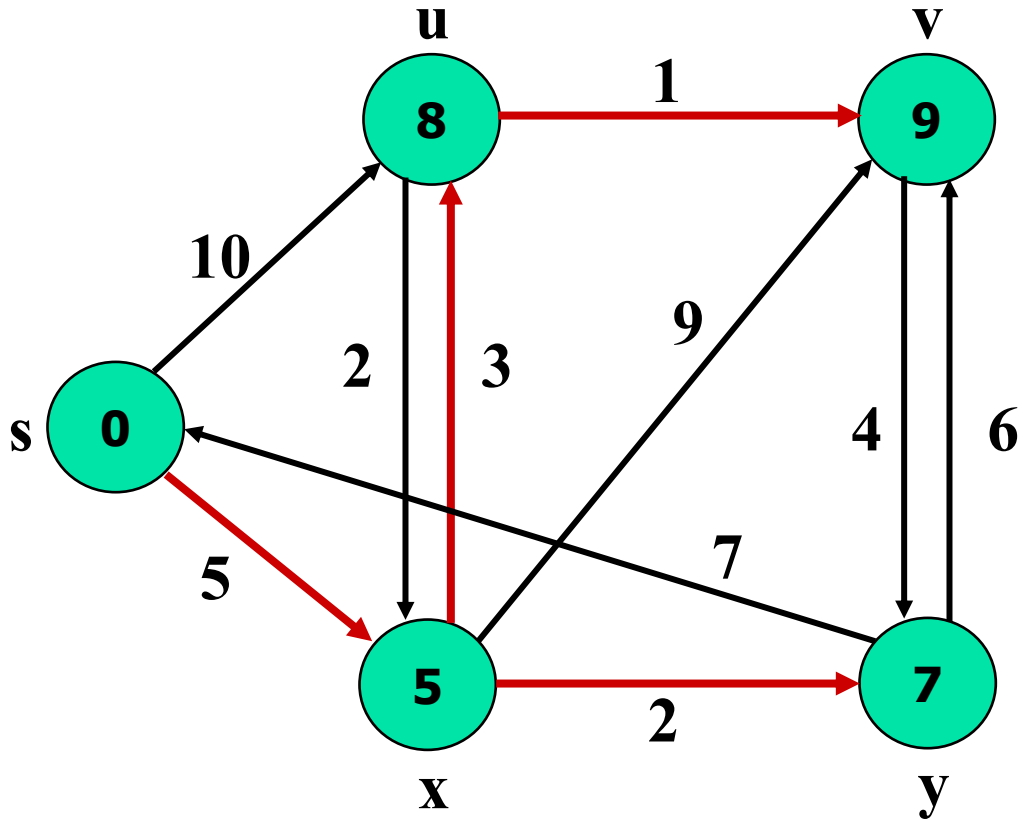
s	u	v	x	y
0	∞	∞	∞	∞
-	10 s	-	5s	-
-	8x	14 x	-	7x
-	8x	13 y	-	-

Example



s	u	v	x	y
0	∞	∞	∞	∞
-	10s	-	5s	-
-	8x	14x	-	7x
-	8x	13y	-	-
-	-	9u	-	-

Example



s	u	v	x	y
0	∞	∞	∞	∞
-	10s	-	5s	-
-	8x	14x	-	7x
-	8x	13y	-	-
-	-	9u	-	-
	8x	9u	5s	7x

Building Routing Tables

to ensure that each node has the routing table showing least-cost node to every other node

- Creation of the states of the links by each node, called the link state packets (LSP)
- Dissemination of LSPs to every other routers, through flooding (efficiently)
- Formation of a shortest path tree for each node
- Calculation of a routing table based on the shortest path tree

Creation of LSP

- LSP data:

E.g. the node ID, the list of links, a sequence number, and age.

- LSP Generation

- When there is a change in the topology of the domain
- On a periodic basis
 - There is no actual need for this type of LSP, normally 60 minutes or 2 hours

Flooding of LSPs

After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbors. The process is called flooding and based on the following:

1. The creating node sends a copy of the LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have.
 - If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP.
 - If it is newer, the node does the following:
 - a.** It discards the old LSP and keeps the new one.
 - b.** It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).

Formation of shortest path tree

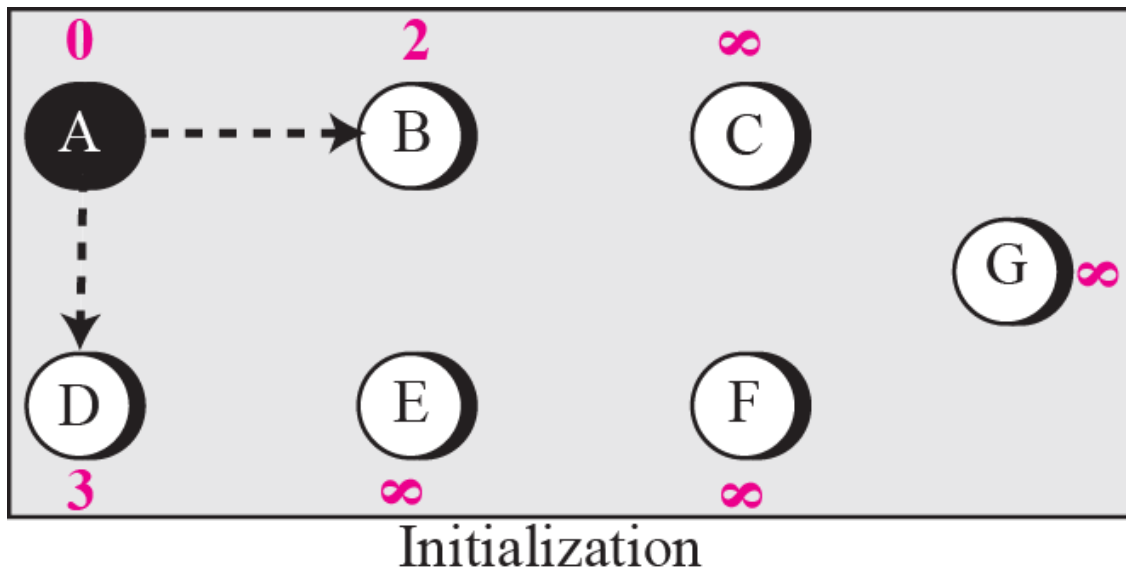
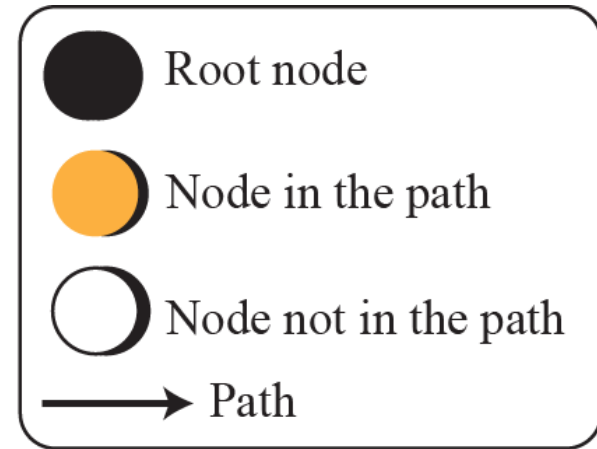
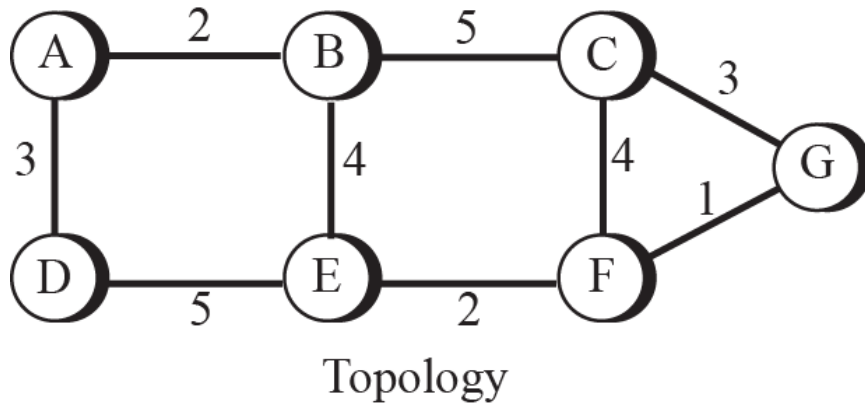
*A shortest path tree is a tree in which the path between **Root** and every other node is the shortest.*

Dijkstra's Algorithm

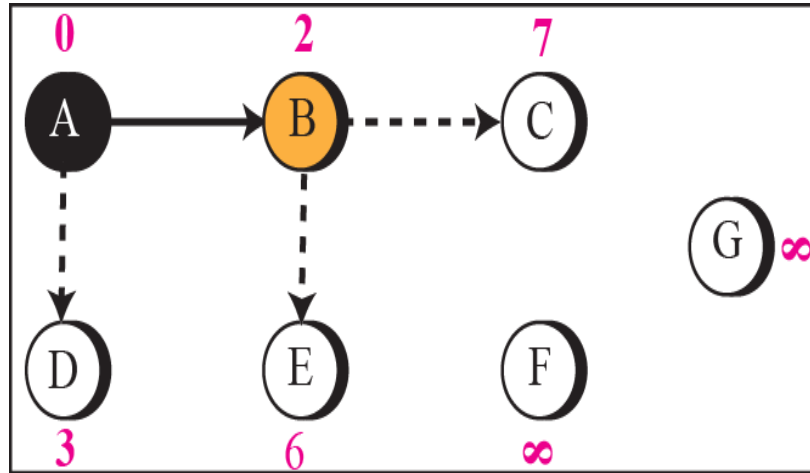
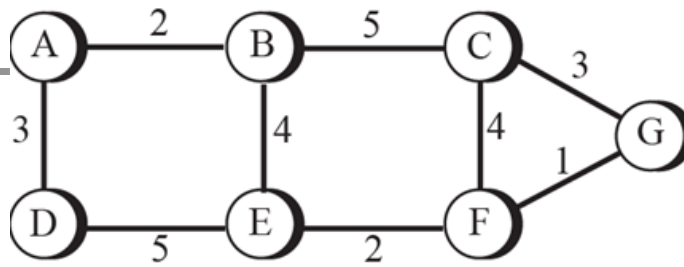
```
1 Dijkstra ( )
2 {
3     // Initialization
4     Path = {s}           // s means self
5     for (i = 1 to N)
6     {
7         if (i is a neighbor of s and  $i \neq s$ )     $D_i = c_{si}$ 
8         if (i is not a neighbor of s)              $D_i = \infty$ 
9     }
10     $D_s = 0$ 
11
12 }
```

```
13      // Iteration
14      Repeat
15      {
16          // Finding the next node to be added
17          Path = Path  $\cup$   $i$    if  $D_i$  is minimum among all remaining nodes
18
19          // Update the shortest distance for the rest
20          for ( $j = 1$  to  $M$ )      //  $M$  number of remaining nodes
21          {
22               $D_j = \text{minimum } (D_j, D_j + c_{ij})$ 
23          }
24      } until (all nodes included in the path,  $M = 0$ )
```

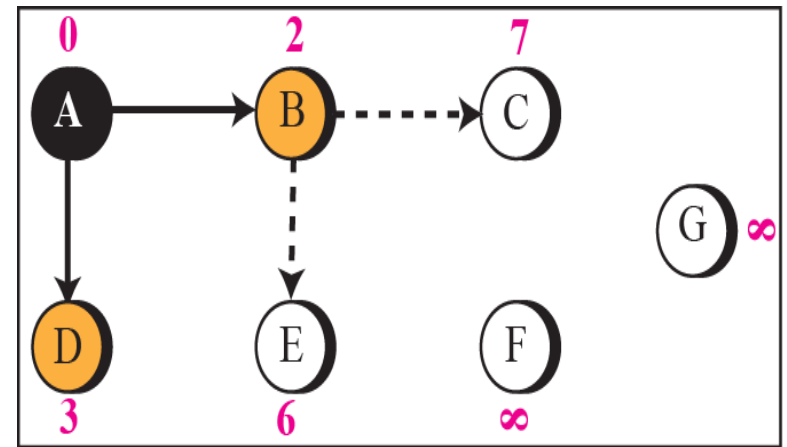

Forming shortest path tree for router A in a graph



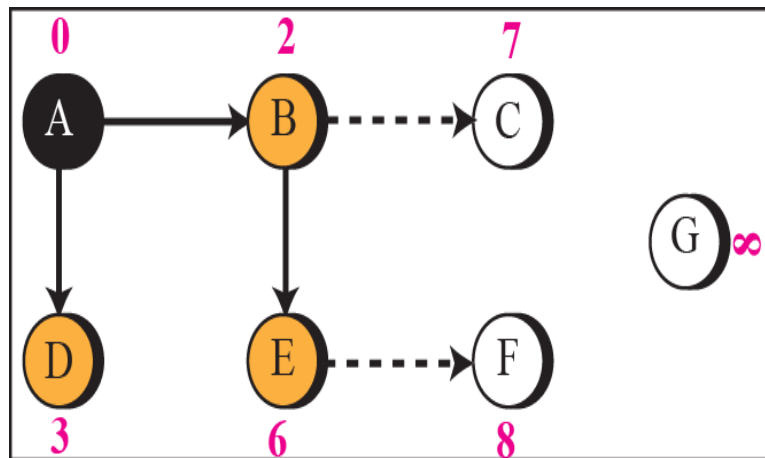
Continued



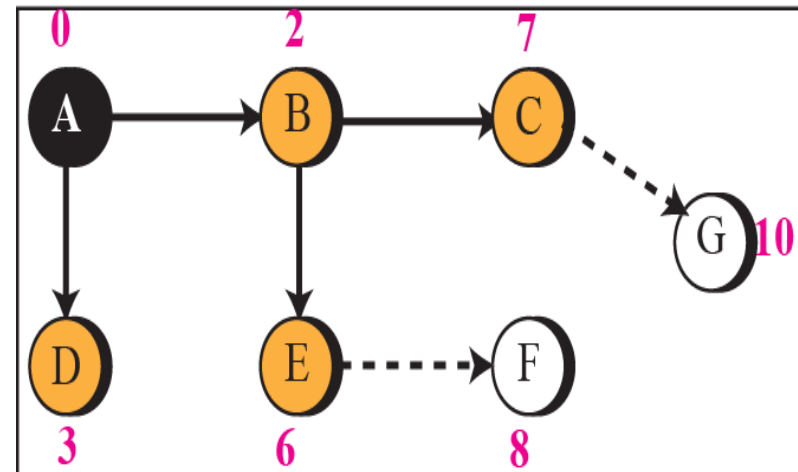
Iteration 1



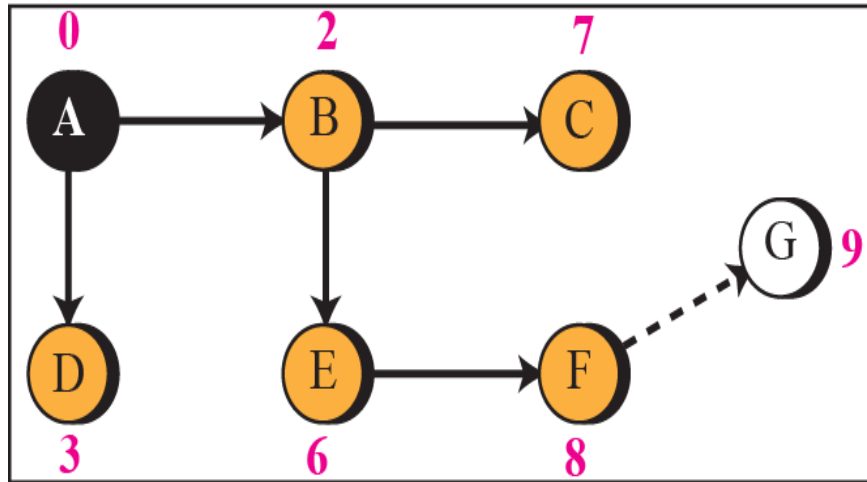
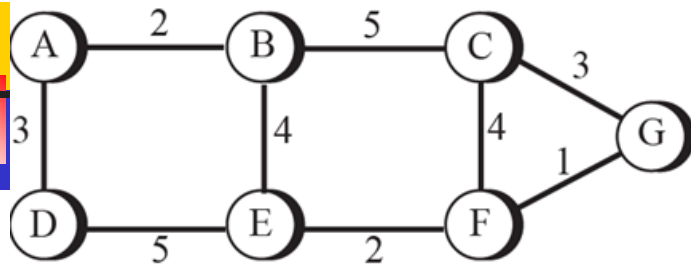
Iteration 2



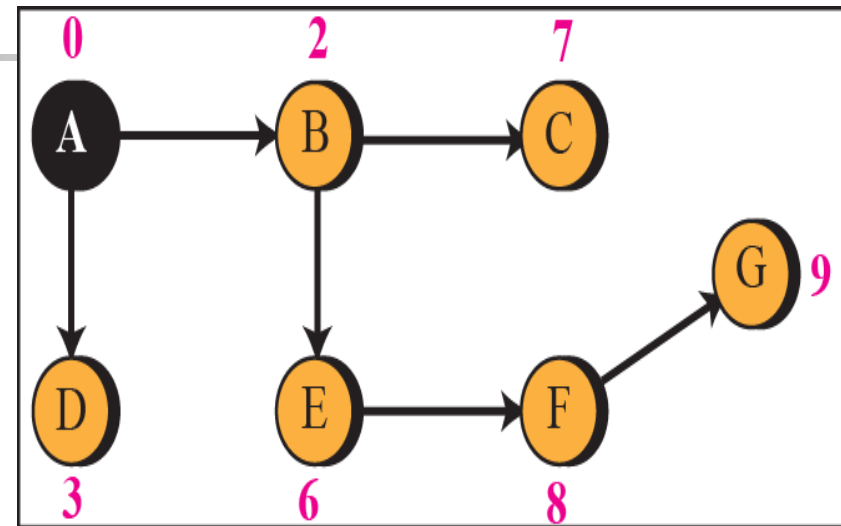
Iteration 3



Iteration 4



Iteration 5



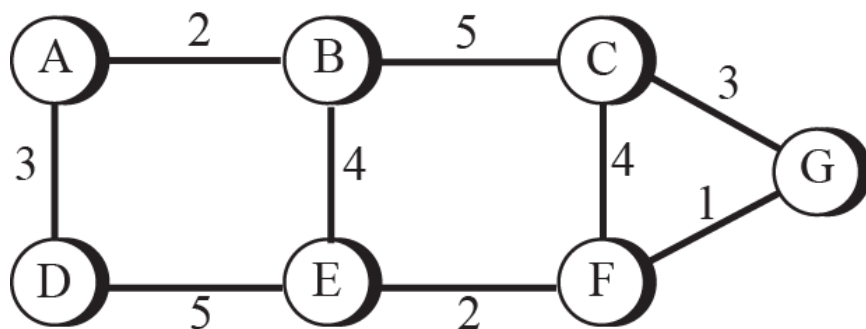
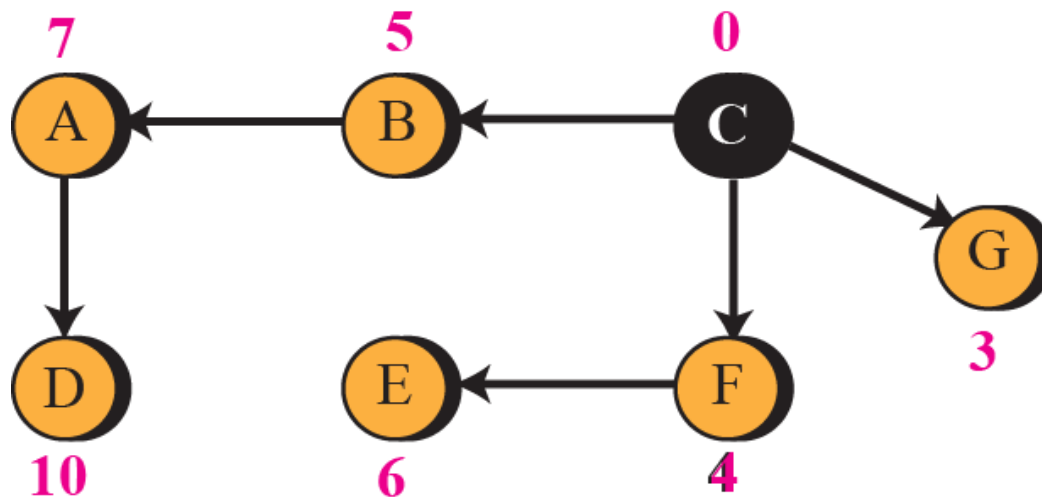
Iteration 6

4 Routing Table for Node A

Destination	Cost	Next Router
A	0	—
B	2	—
C	7	B
D	3	—
E	6	B
F	8	B
G	9	B

Example

the shortest path tree for each node is different,
the shortest path tree as seen by node C



Topology

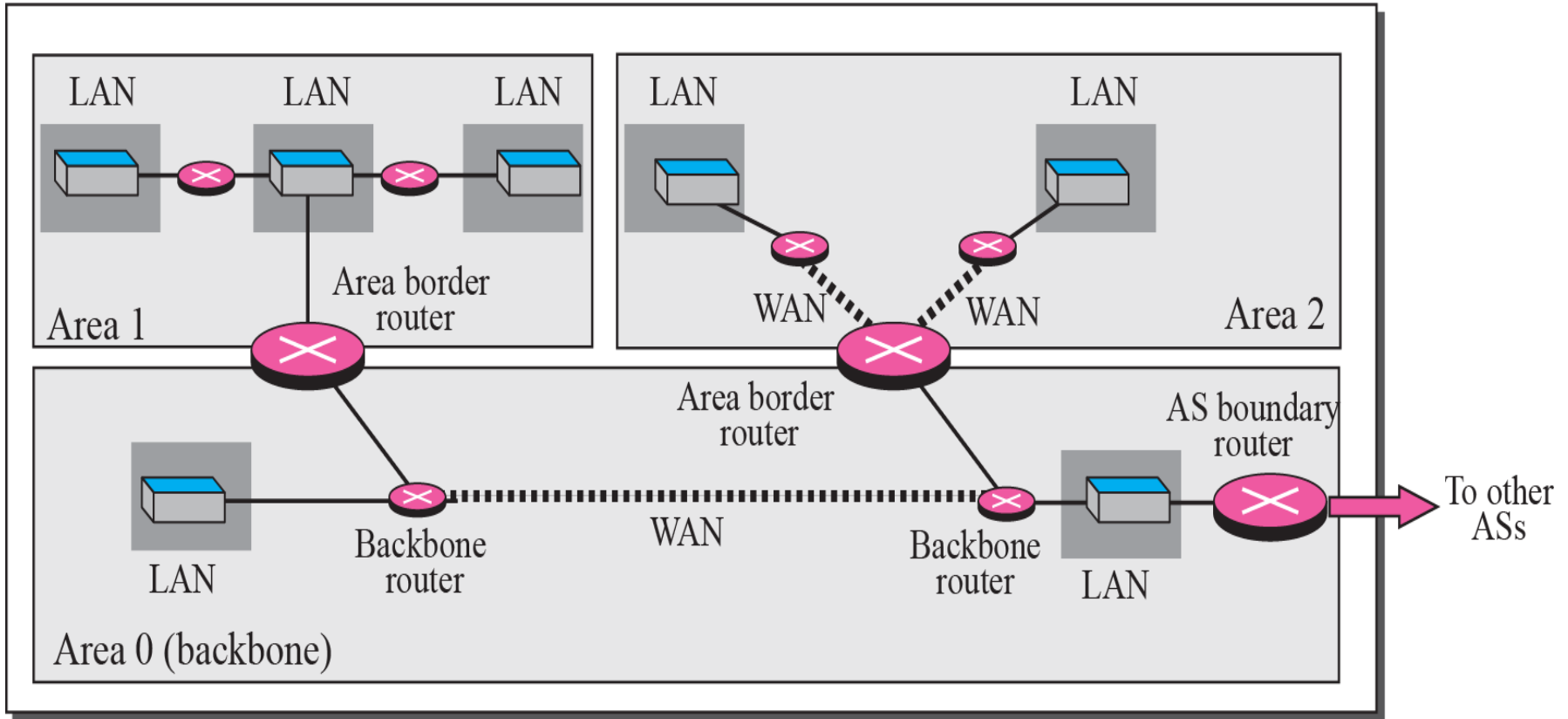
Create for B
Shortest Path ?
Routing Table ?

The Open Shortest Path First (OSPF) protocol is an intra-domain routing protocol based on link state routing.

Its domain is also an autonomous system.

Areas in an autonomous system

Autonomous System (AS)



An area is a collection of networks, hosts, and routers all contained within an autonomous system. Each Area has area identification, Area identification for a backbone is zero

Types of links

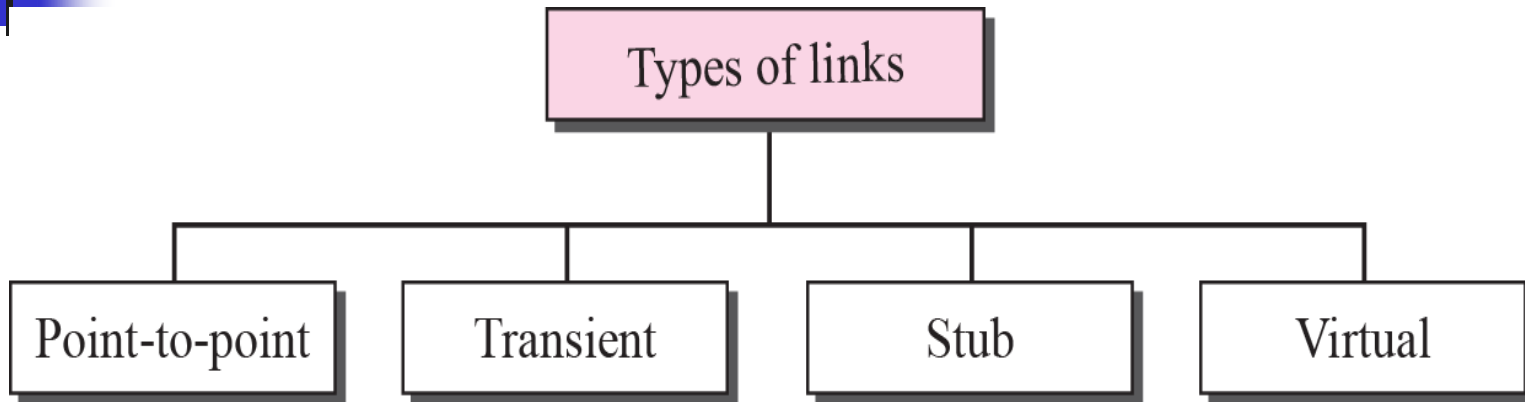


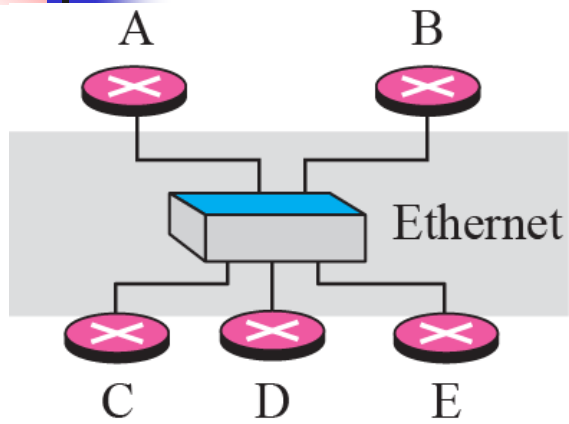
Figure 11.23 *Point-to-point link*



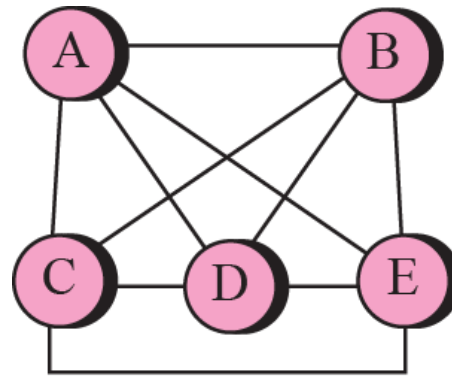
a. Point-to-point network



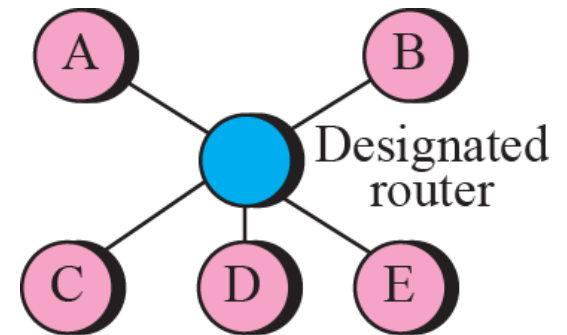
b. Representation



a. Transient network

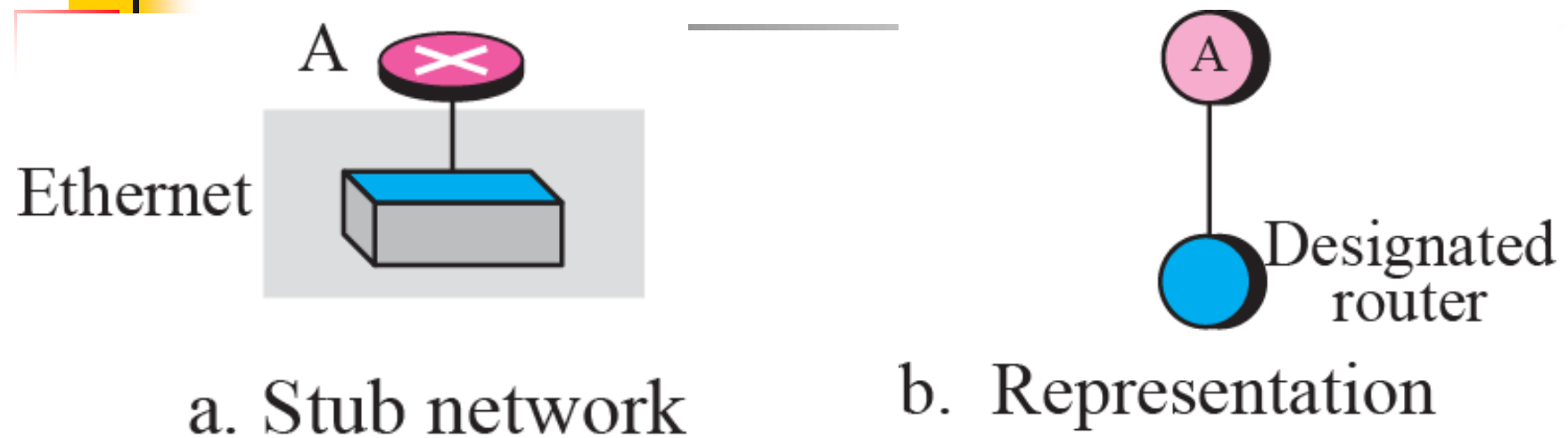


b. Unrealistic



c. Realistic

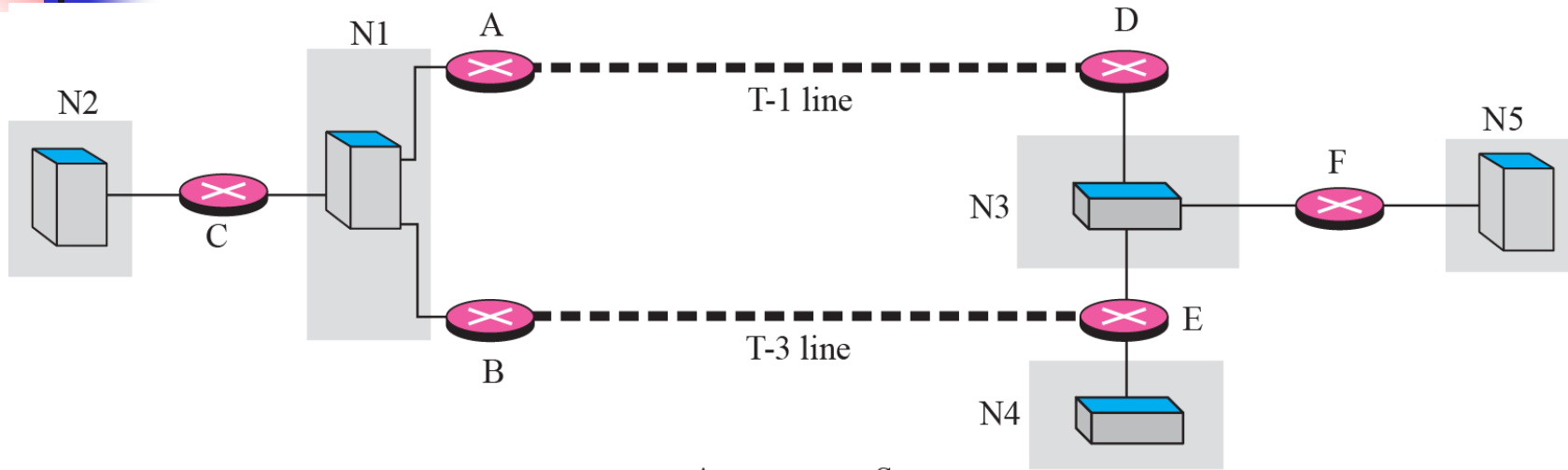
Stub link



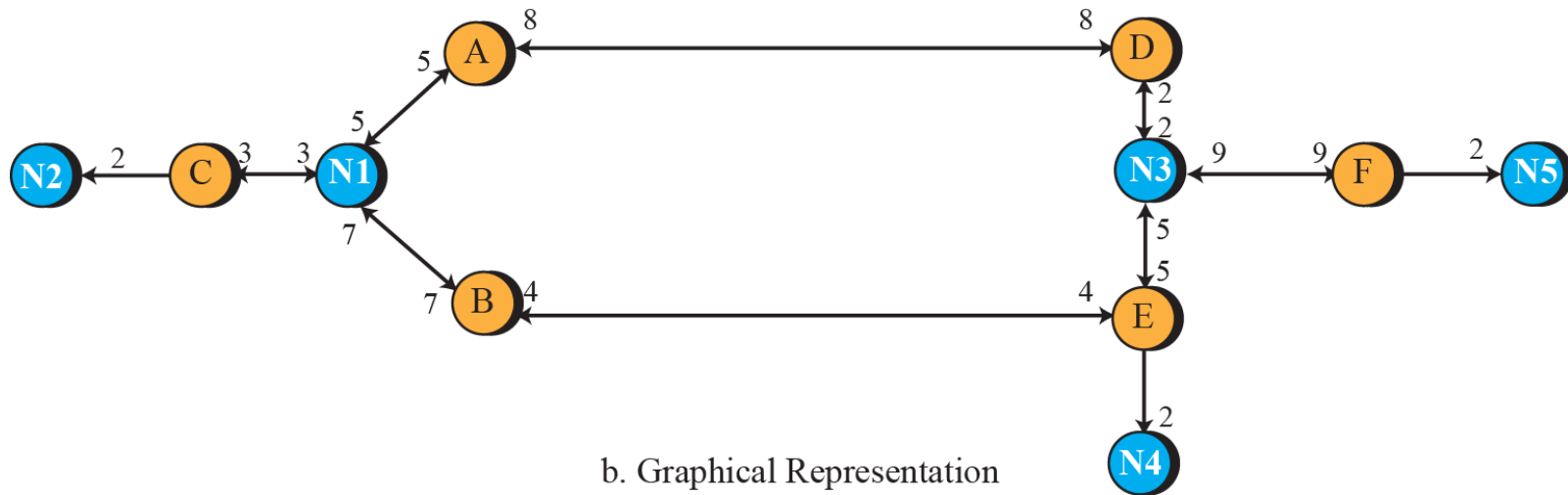
■ Virtual link

- If, due to some problem, the connectivity between a backbone and an area is broken, a virtual link between routers must be created by the administration to allow continuity of the functions of the backbone as the primary area

Example of an AS and its graphical representation in OSPF

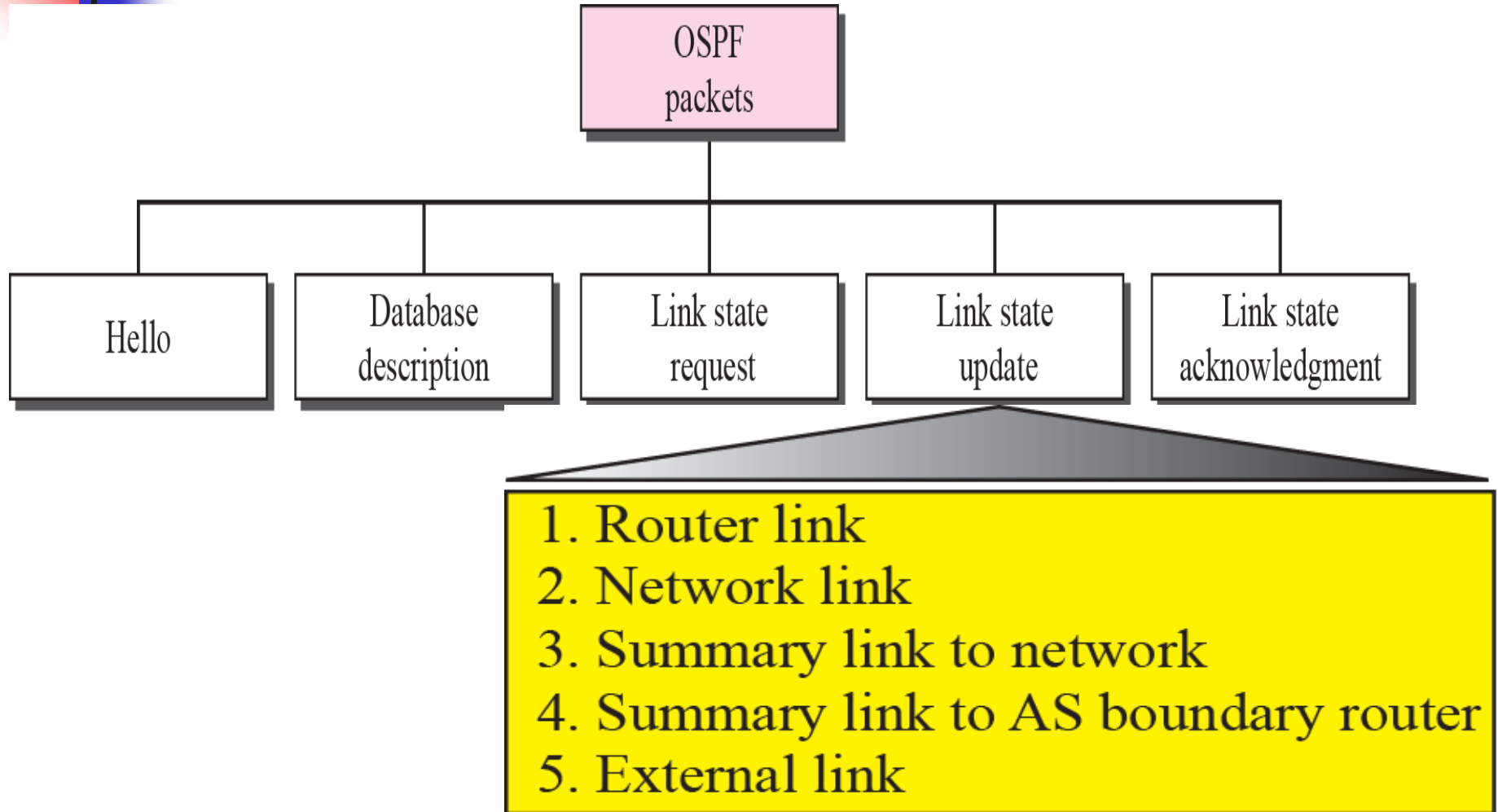


a. Autonomous System

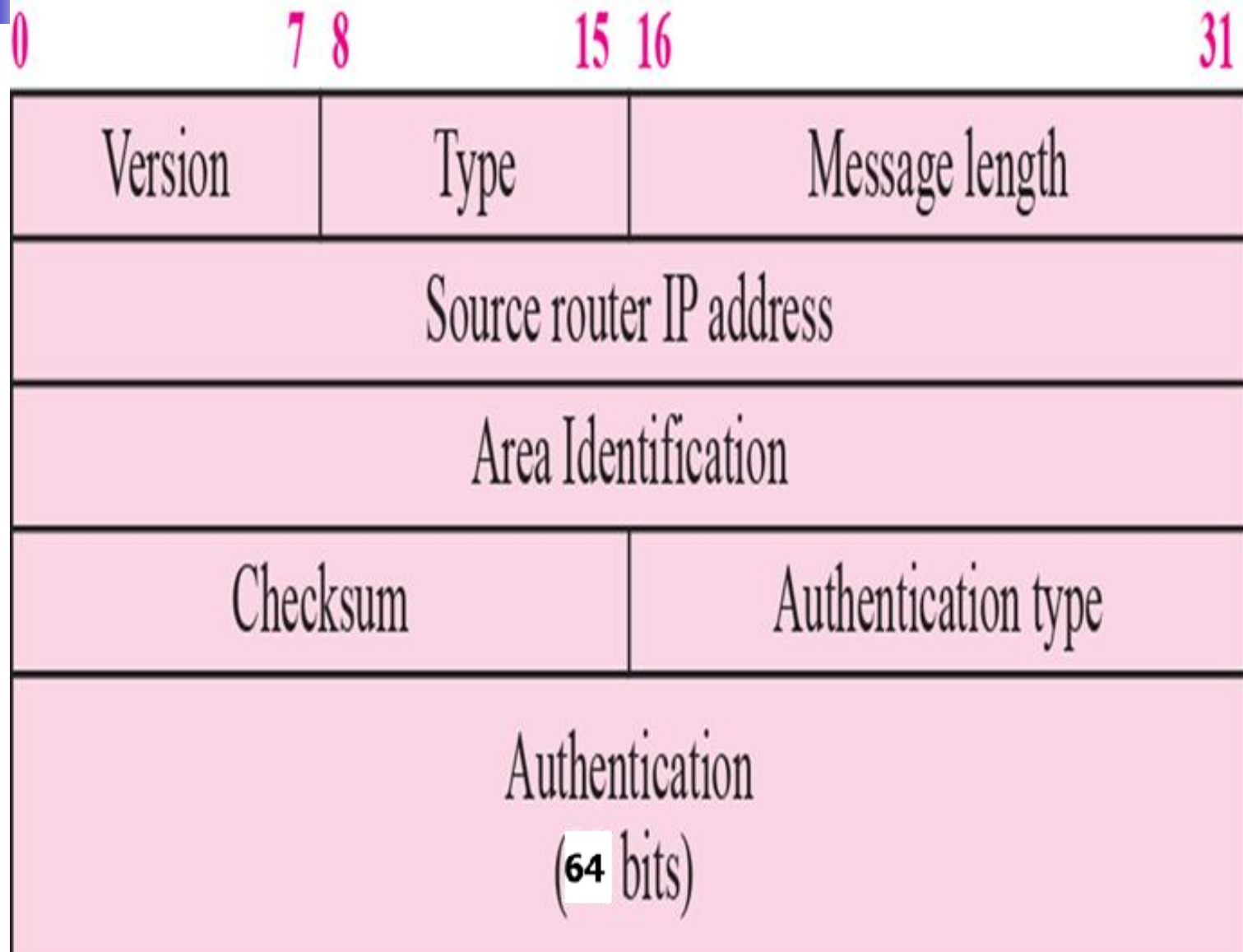


b. Graphical Representation

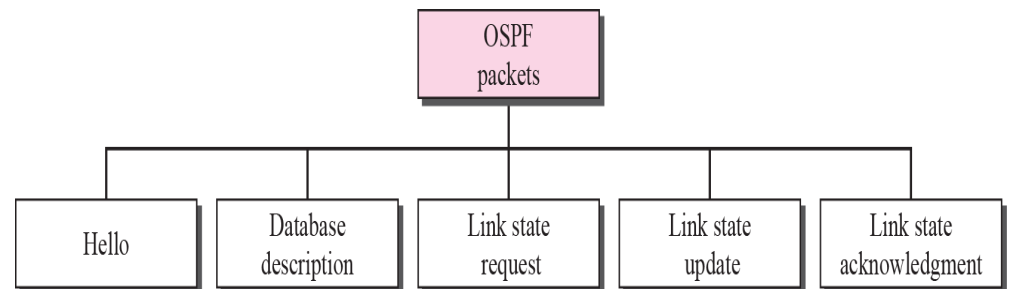
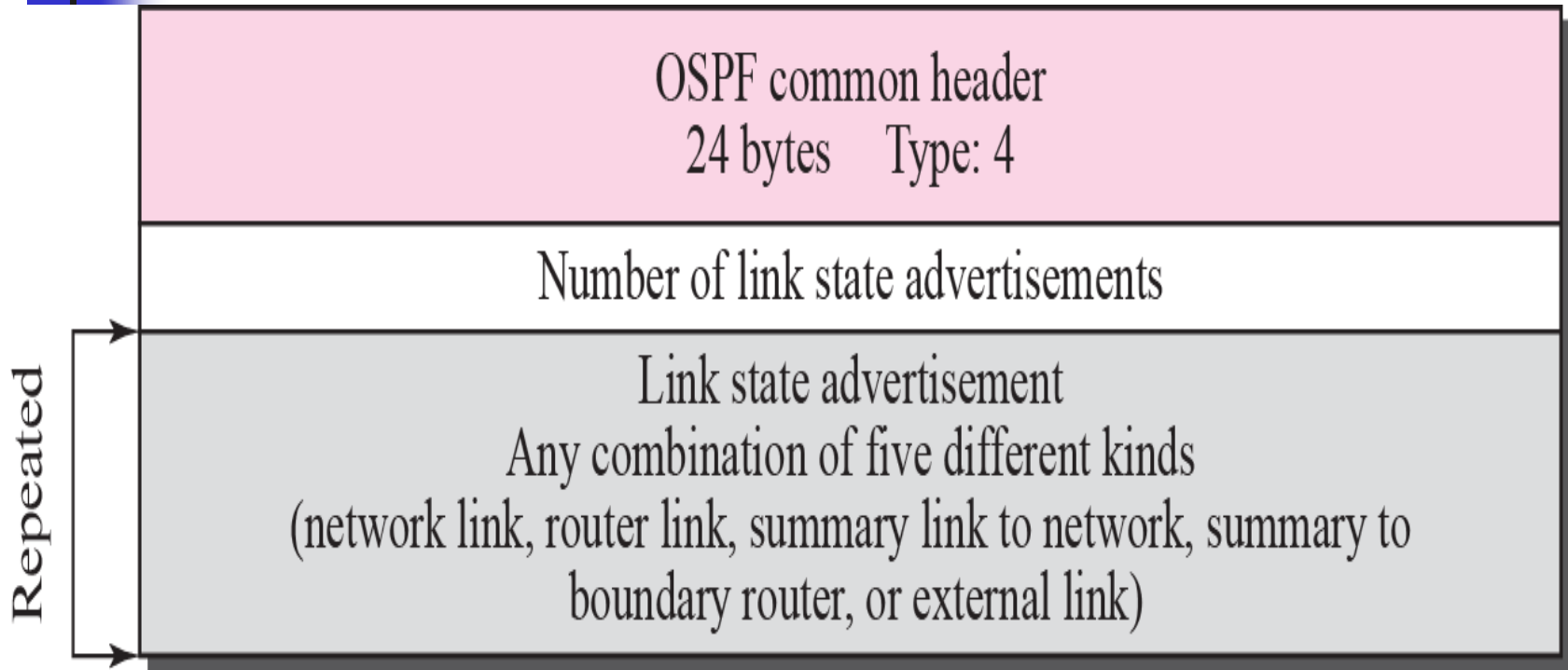
Types of OSPF packet



- *OSPF common header*



Link state update packet



LSA general header

Link state age	Reserved	E	T	Link state type
Link state ID				
Advertising router				
Link state sequence number				
Link state checksum	Length			

Need for the fields “sequence number” and “age” in packets ?

Sequence number keeps flooding in check (ensures that the same packet cannot be flooded more than once by every node).

Age prevents ambiguities that arise from wrapping around of sequence numbers. (*like timestamping feature to distinguish*)

LSA General Header

- Link state age
 - When a router creates the message, the value of this field is 0
 - When each successive router forwards this message, it estimates the transit time and adds it to the cumulative value of this field

Link state age	Reserved	E	T	Link state type
Link state ID				
Advertising router				
Link state sequence number				
Link state checksum		Length		

- E flag
 - If this flag is set to 1, it means the area is a stub area (an area that is connected to the backbone area by only one path)
- T flag
 - If this flag is set to 1, it means the router can handle multiple types of services

LS Type and LS ID

Link state age	Reserved	E	T	Link state type
Link state ID				
Advertising router				
Link state sequence number				
Link state checksum		Length		

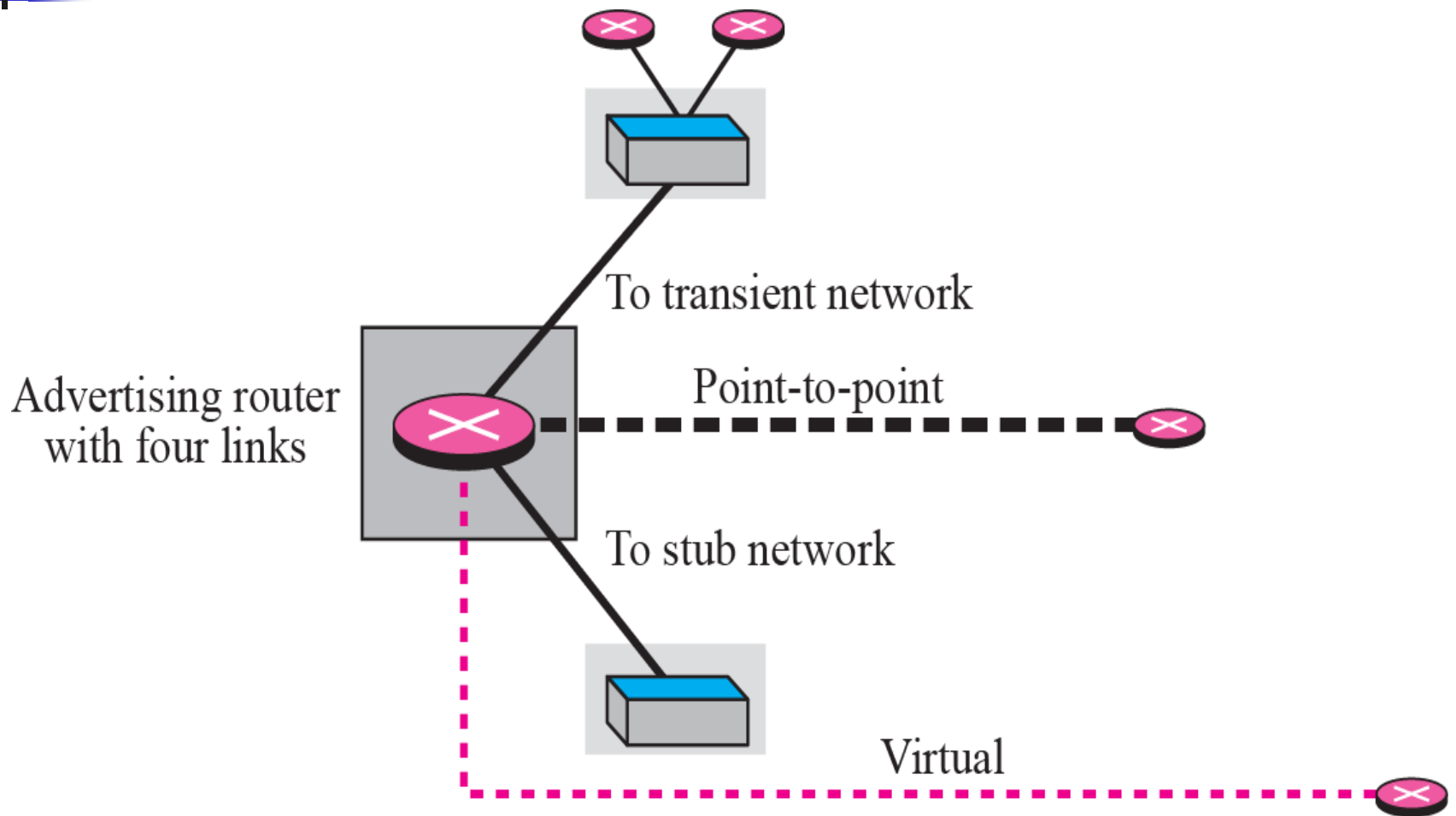
Link state type	Link state ID
Router link	IP address of the router
Network link	IP address of the designated router
Summary link to network	Address of the network
Summary link to AS boundary	IP address of the boundary router
External link	Address of the external network

LSA General Header

- Advertising router
 - The IP address of the router advertising this message
- Link state sequence number
 - A sequence number assigned to each link state update message

Link state age	Reserved	E	T	Link state type
Link state ID				
Advertising router				
Link state sequence number				
Link state checksum		Length		

Router link



Router link LSA

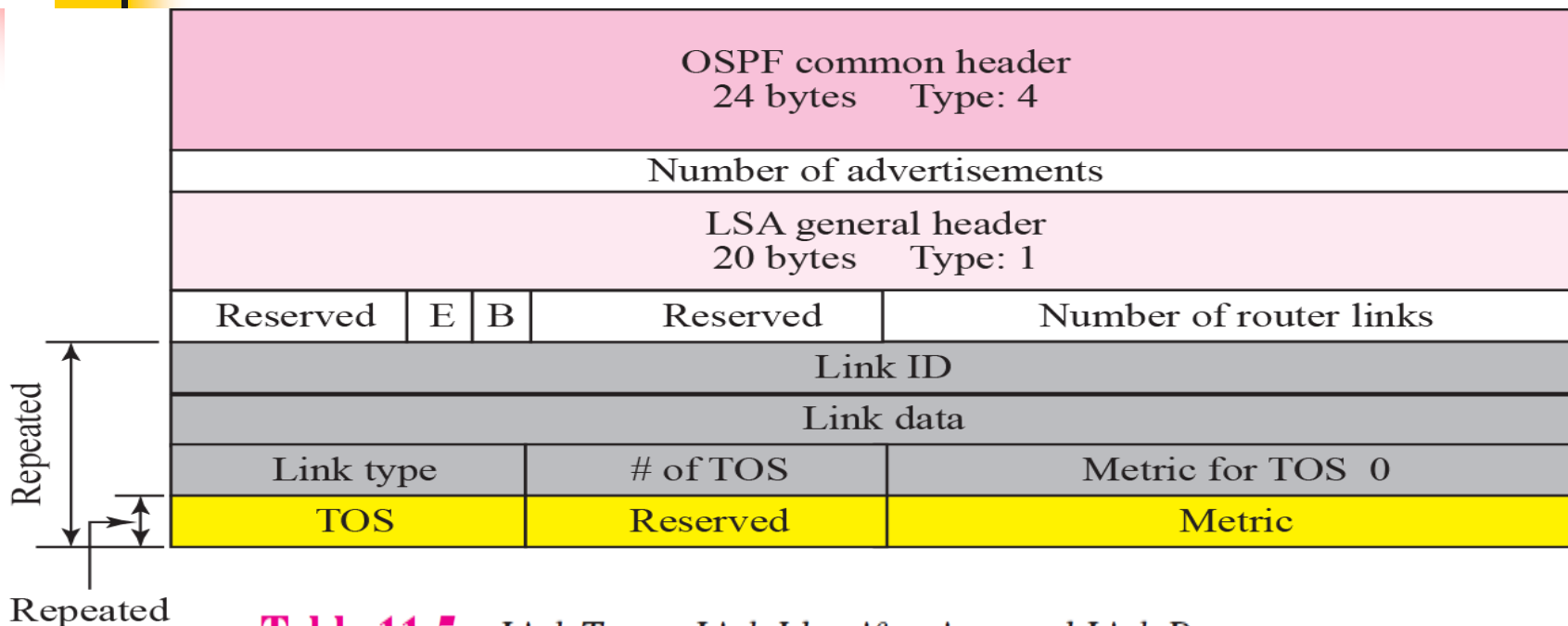
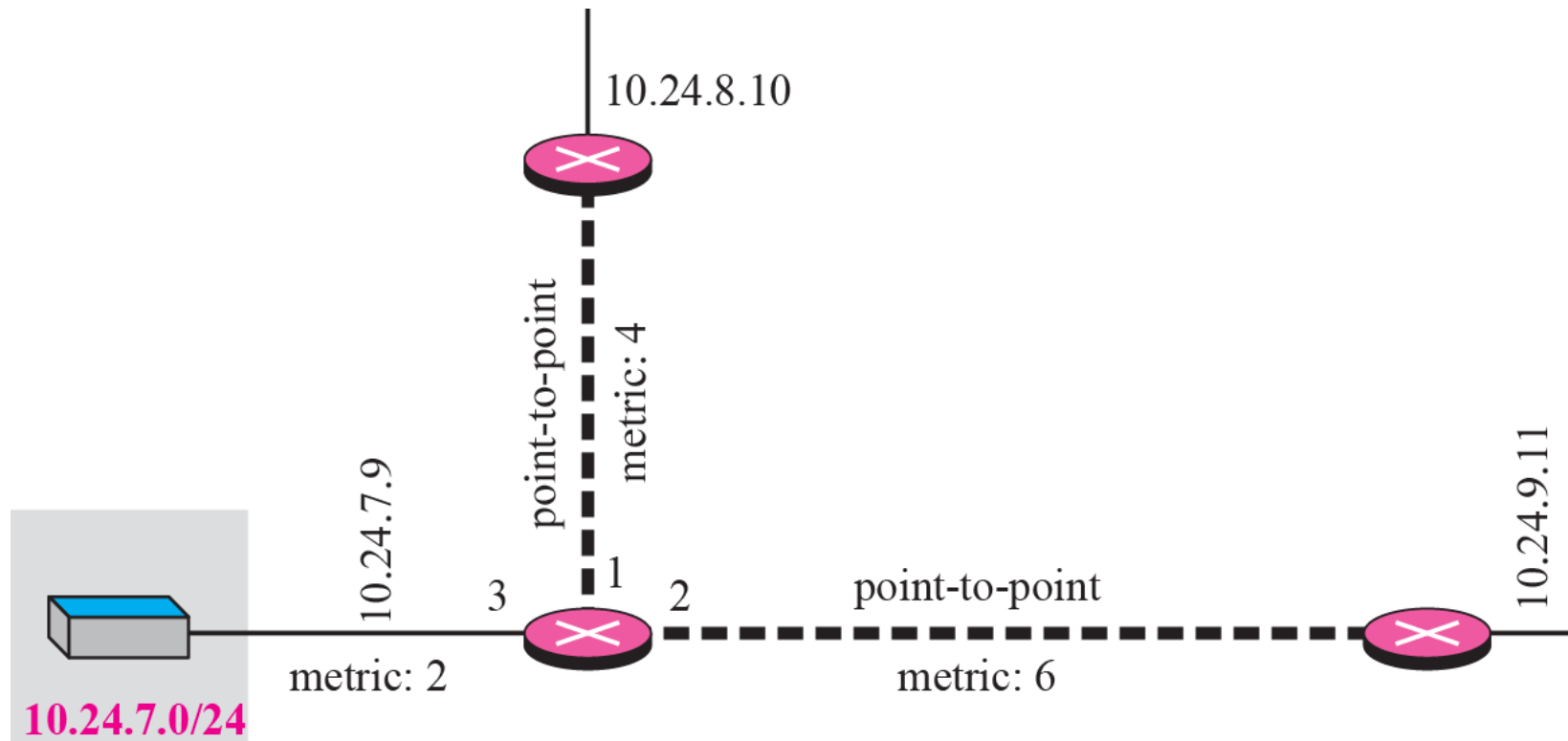


Table 11.5 *Link Types, Link Identification, and Link Data*

<i>Link Type</i>	<i>Link Identification</i>	<i>Link Data</i>
Type 1: Point-to-point	Address of neighbor router	Interface number
Type 2: Transient	Address of designated router	Router address
Type 3: Stub	Network address	Network mask
Type 4: Virtual	Address of neighbor router	Router address

Example

shows the router link LSA sent by router 10.24.7.9 in Figure



Solution

This router has three links:

two of type 1 (point-to-point) and
one of type 3 (stub network).

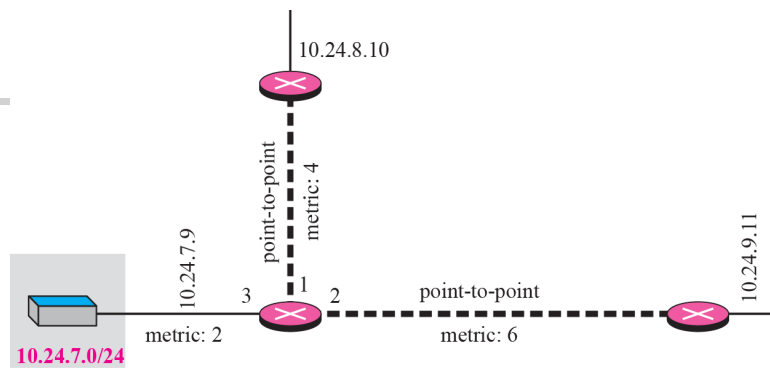
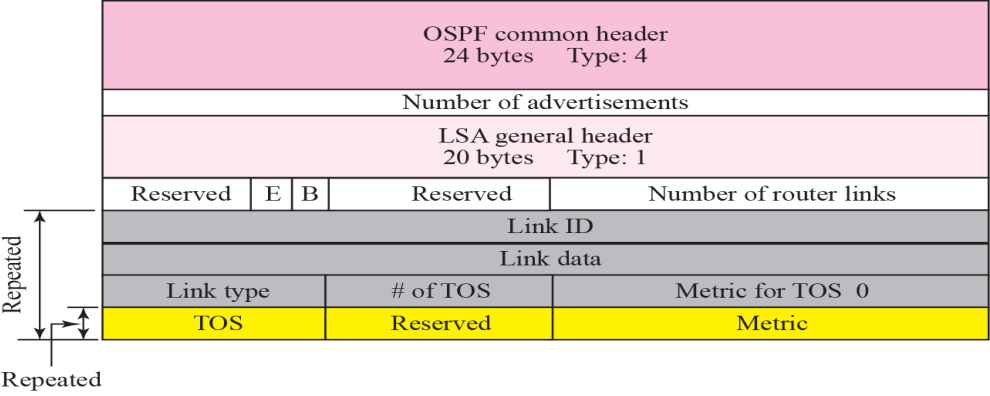
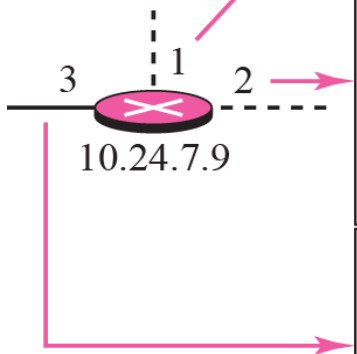


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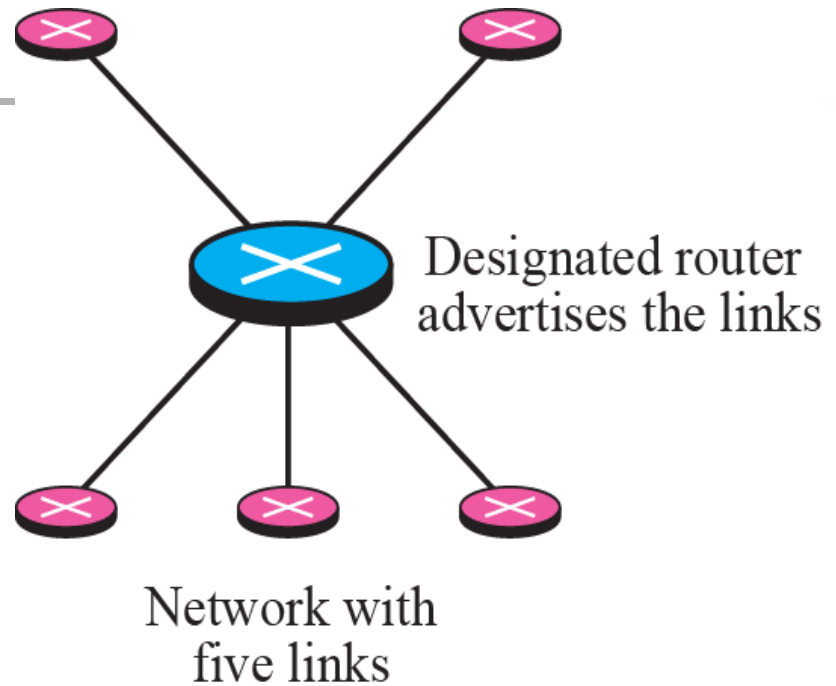
Solution



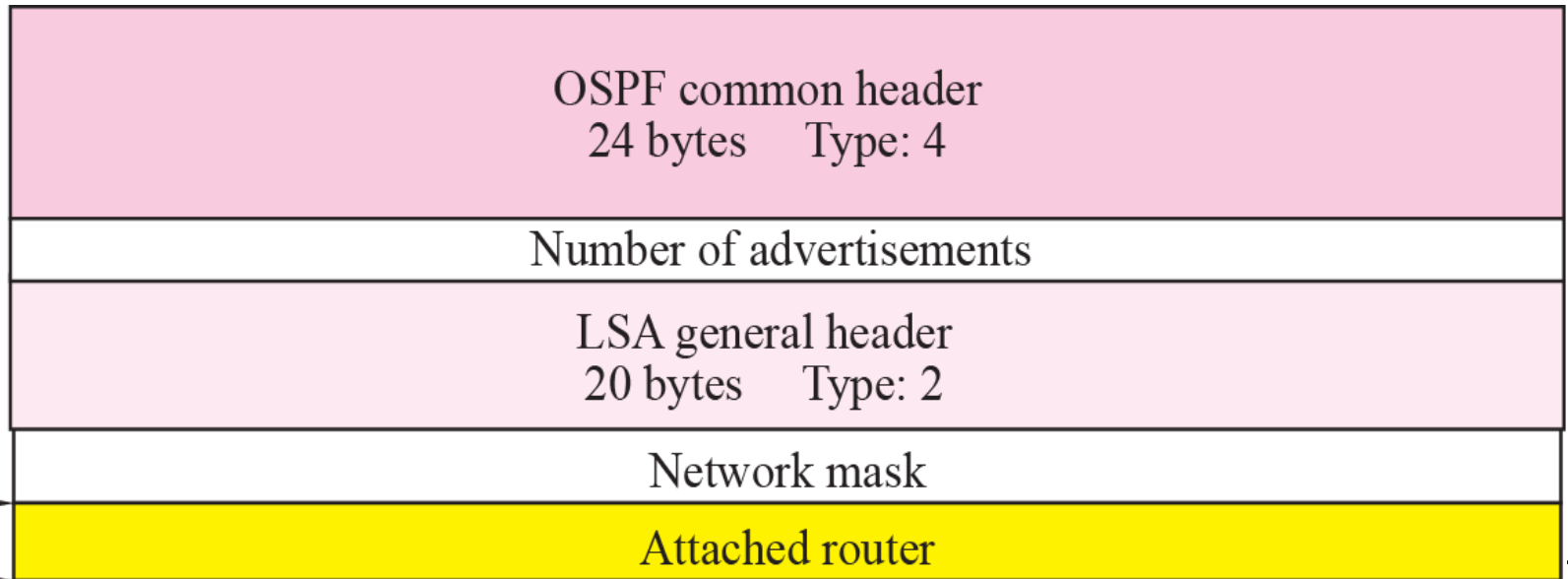
OSPF common header Type: 4		
Number of advertisements: 1		
LSA general header Type: 1		
		Number of links: 3
10.24.8.10		
1		
1		
		4
10.24.9.11		
2		
1		
		6
10.24.7.0		
255.255.255.0		
3		
		2

One router link advertisement

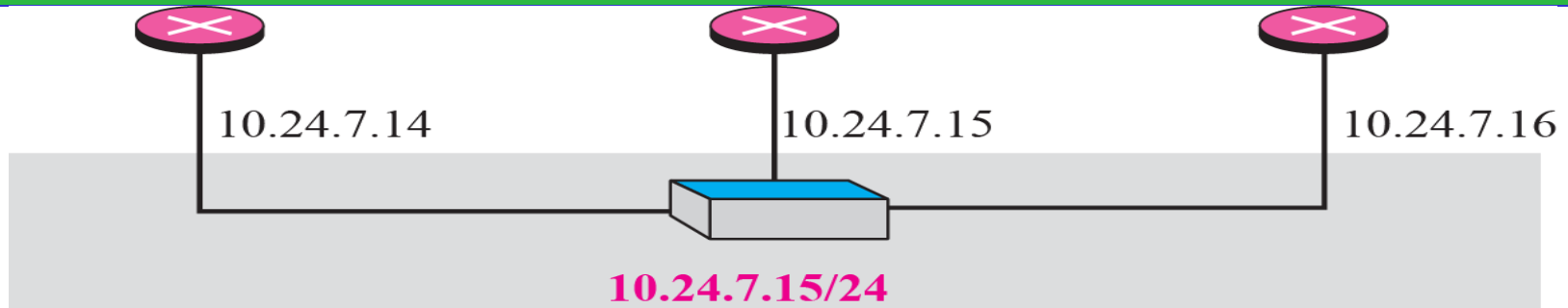
Network link



Network link advertisement format



Example Give the network link LSA in Figure



Solution

The network for which the network link advertises has three routers attached. The LSA shows the mask and the router addresses.

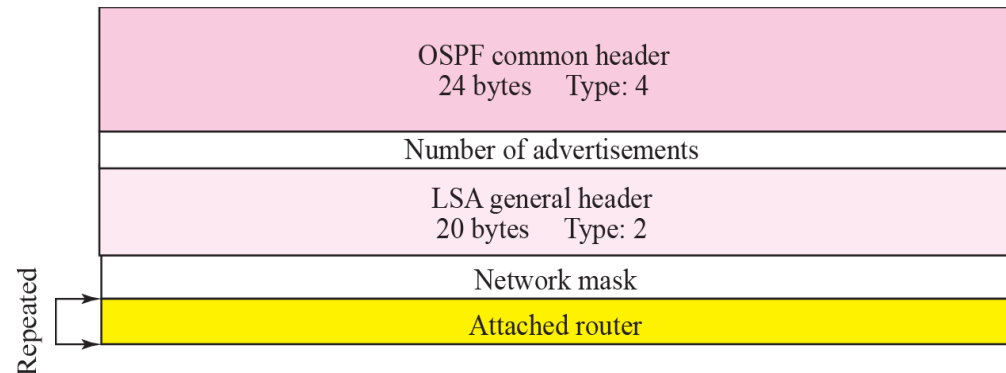
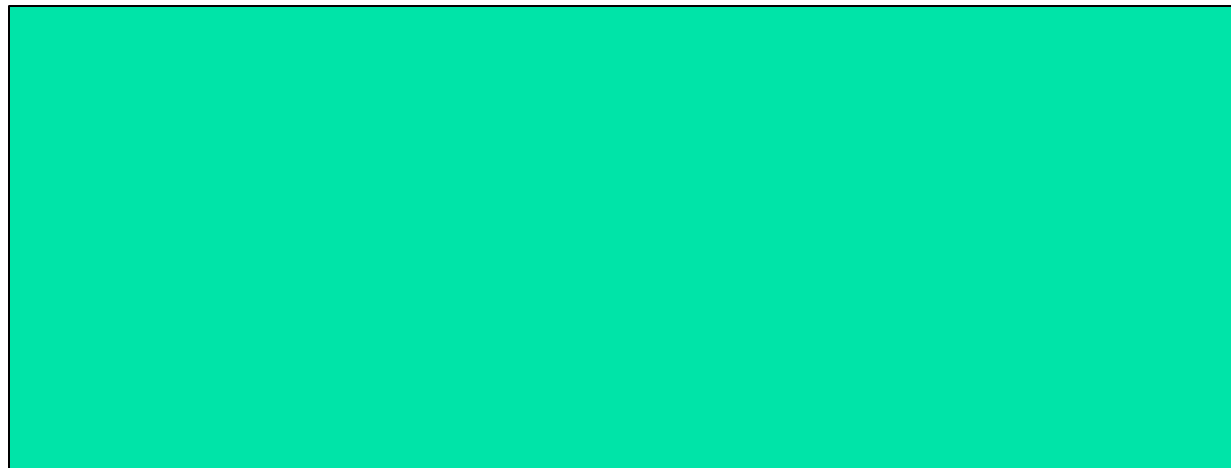
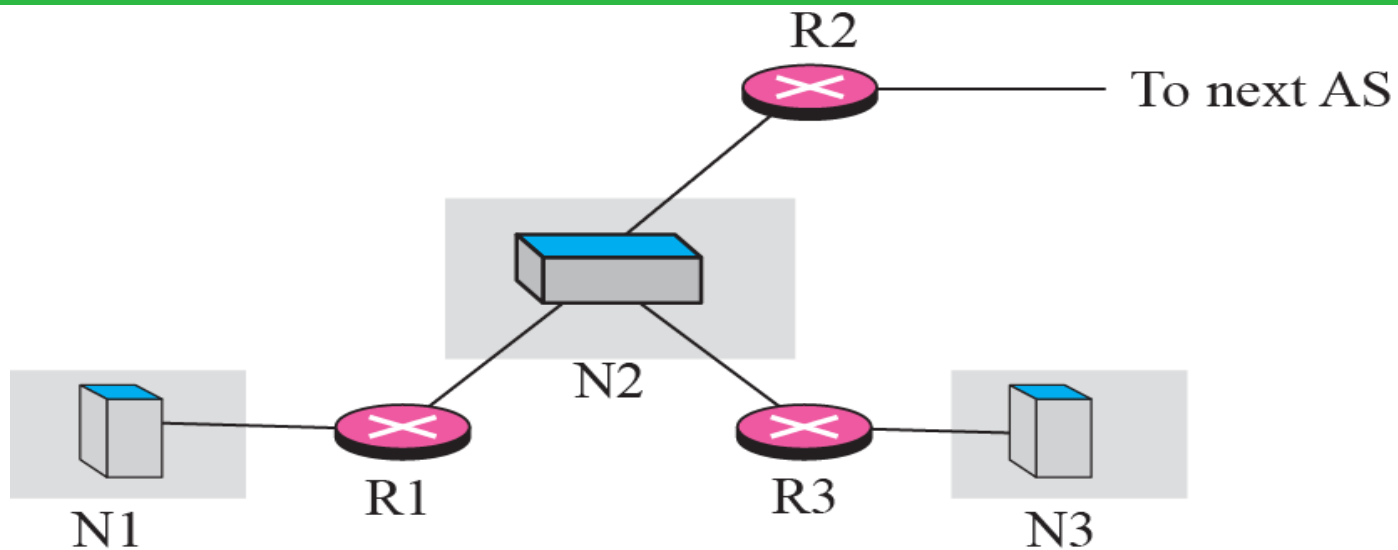


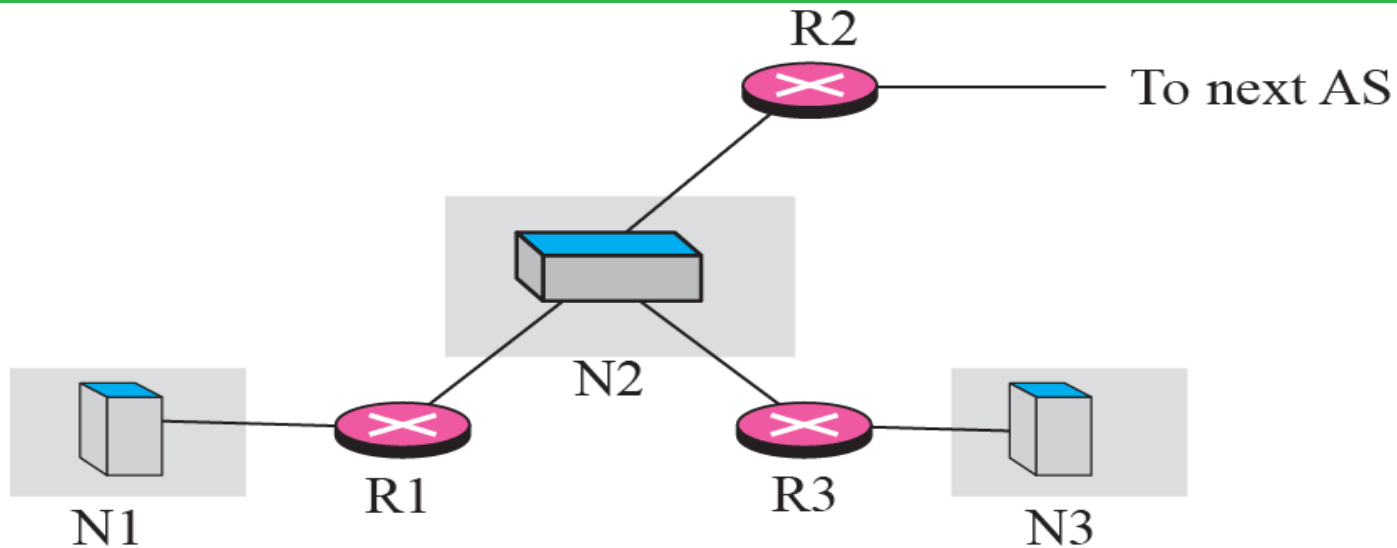
Figure shows the network link LSA.

OSPF common header	Type: 4
Number of advertisements: 1	
LSA general header	Type: 2
255.255.255.0	
10.24.7.14	
10.24.7.15	
10.24.7.16	

Example In Figure , which router(s) sends out router link LSAs?



Example In Figure , which router(s) sends out router link LSAs?
LSAs?

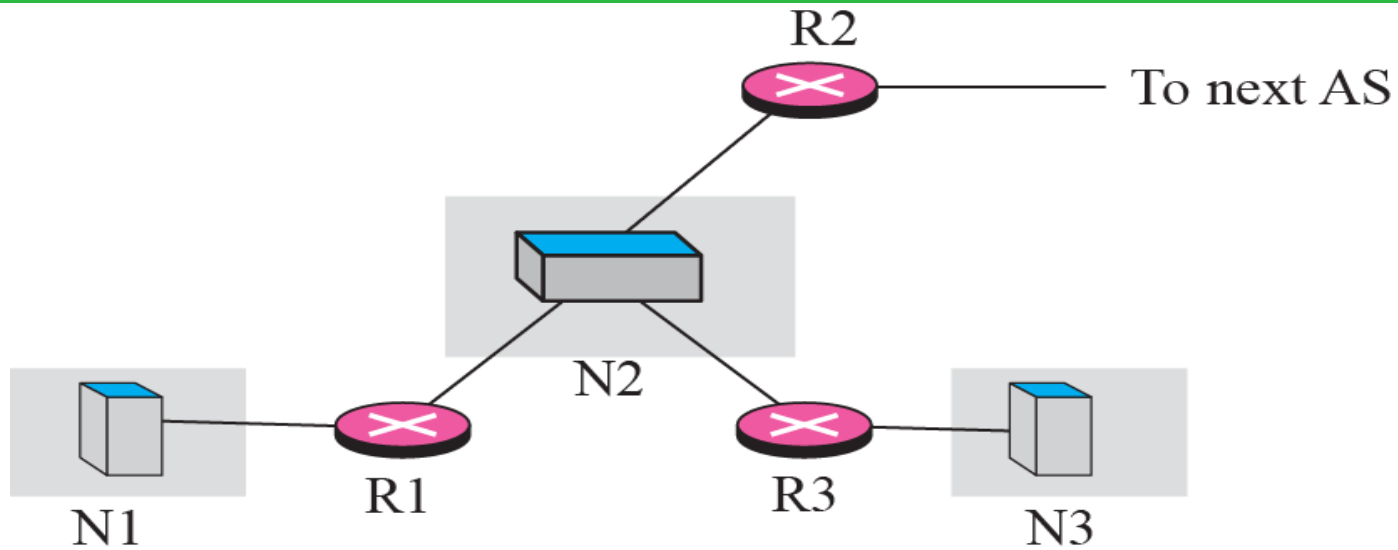


Solution

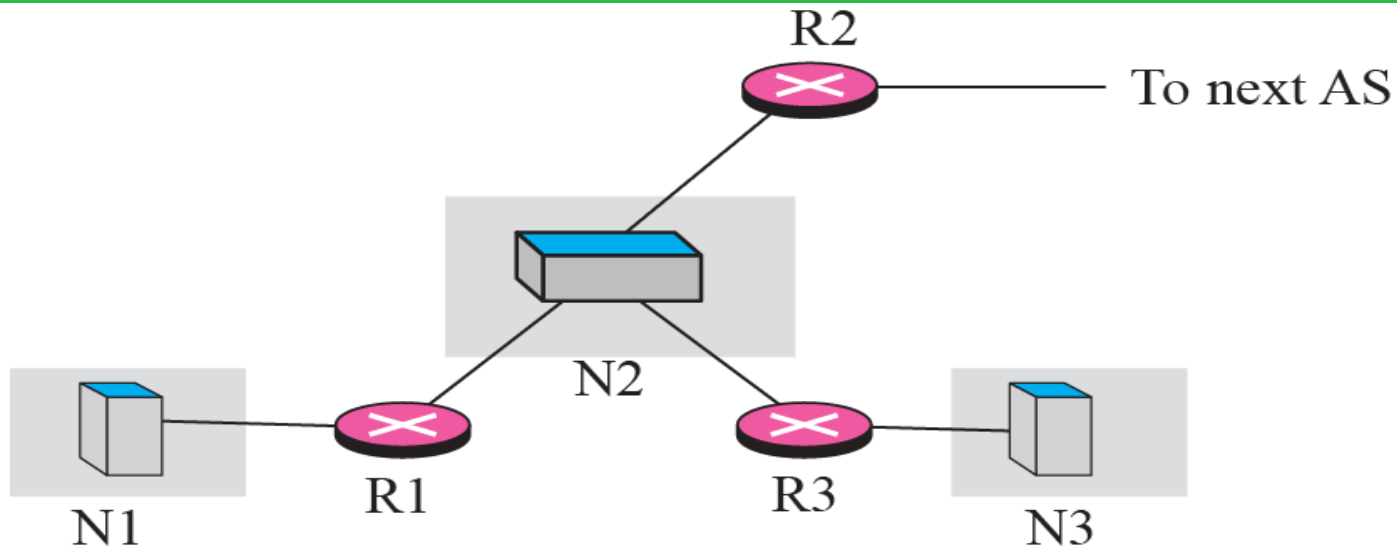
All routers advertise router link LSAs.

- a.** R1 has two links, N1 and N2.
- b.** R2 has one link, N2.
- c.** R3 has two links, N2 and N3.

Example In Figure , which router(s) sends out network link LSAs?



Example In Figure , which router(s) sends out network link LSAs?

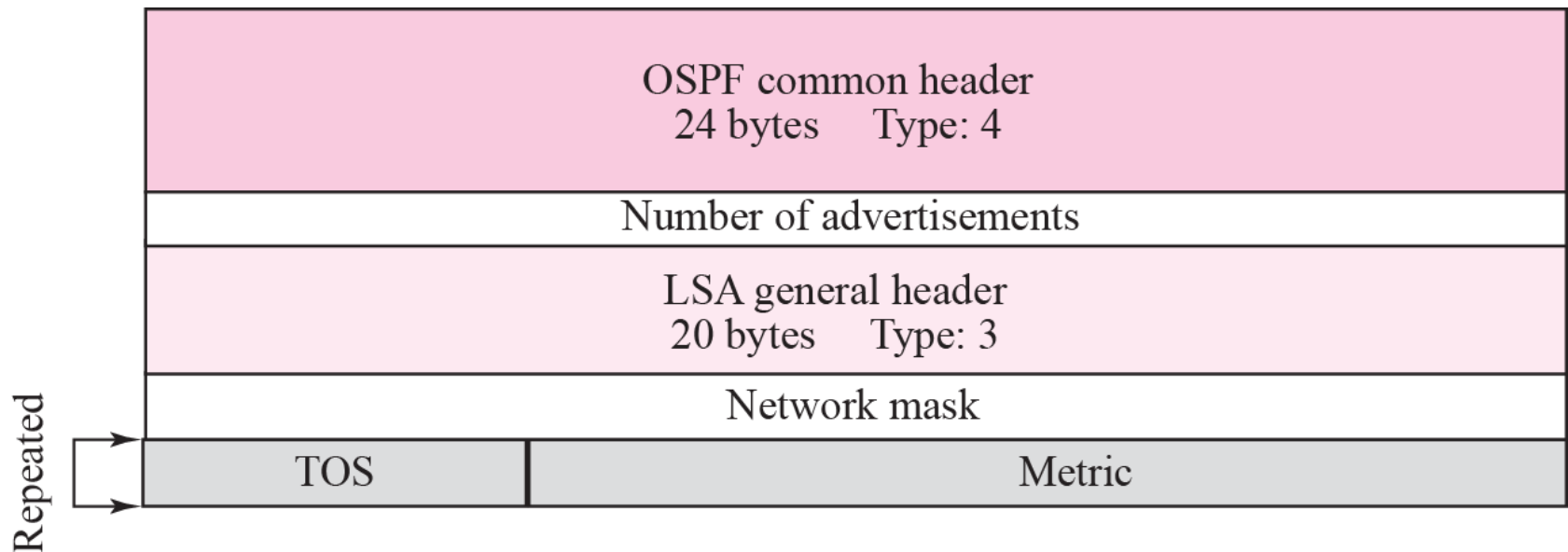
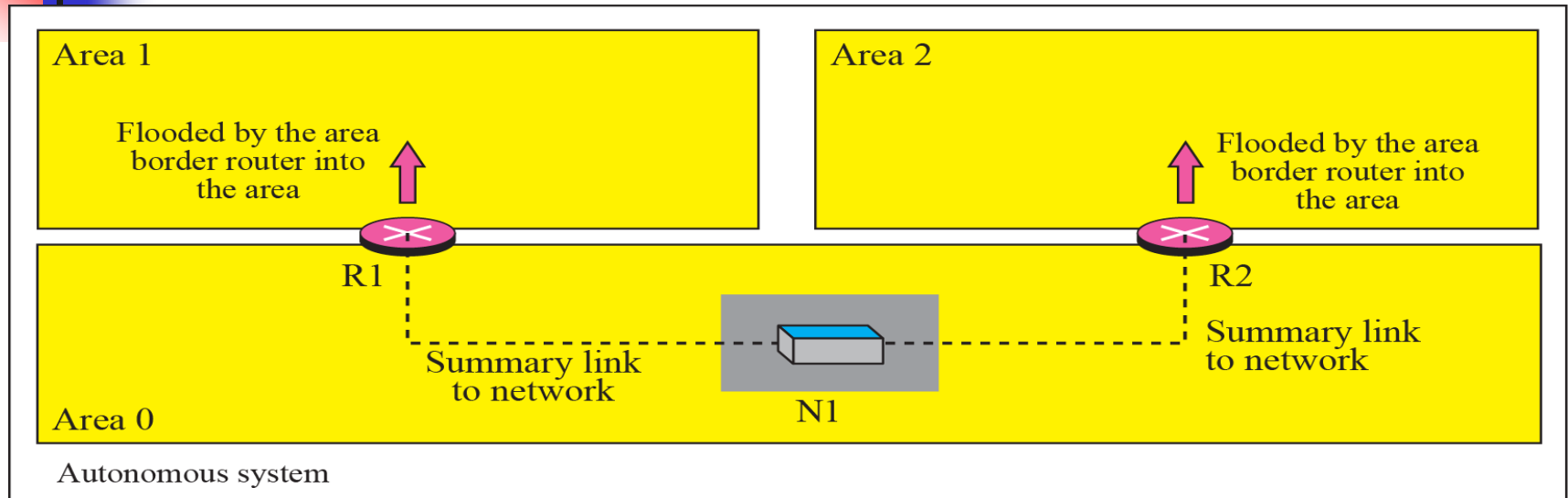


Solution

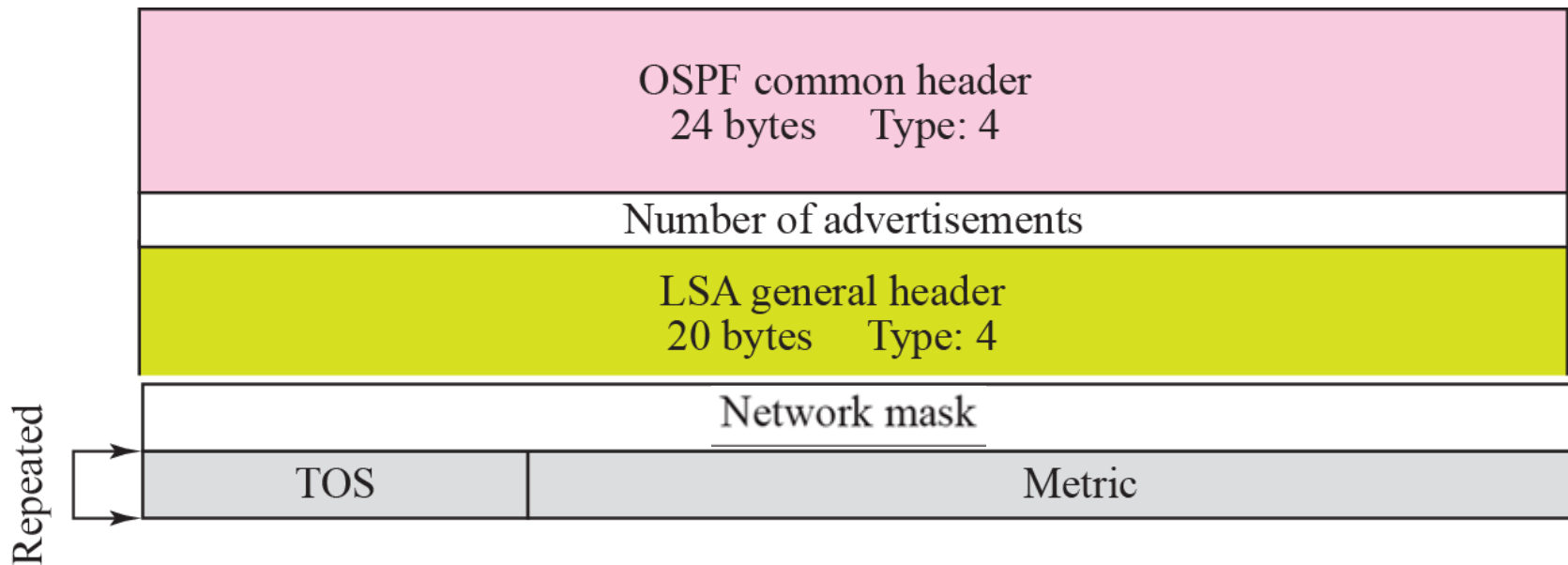
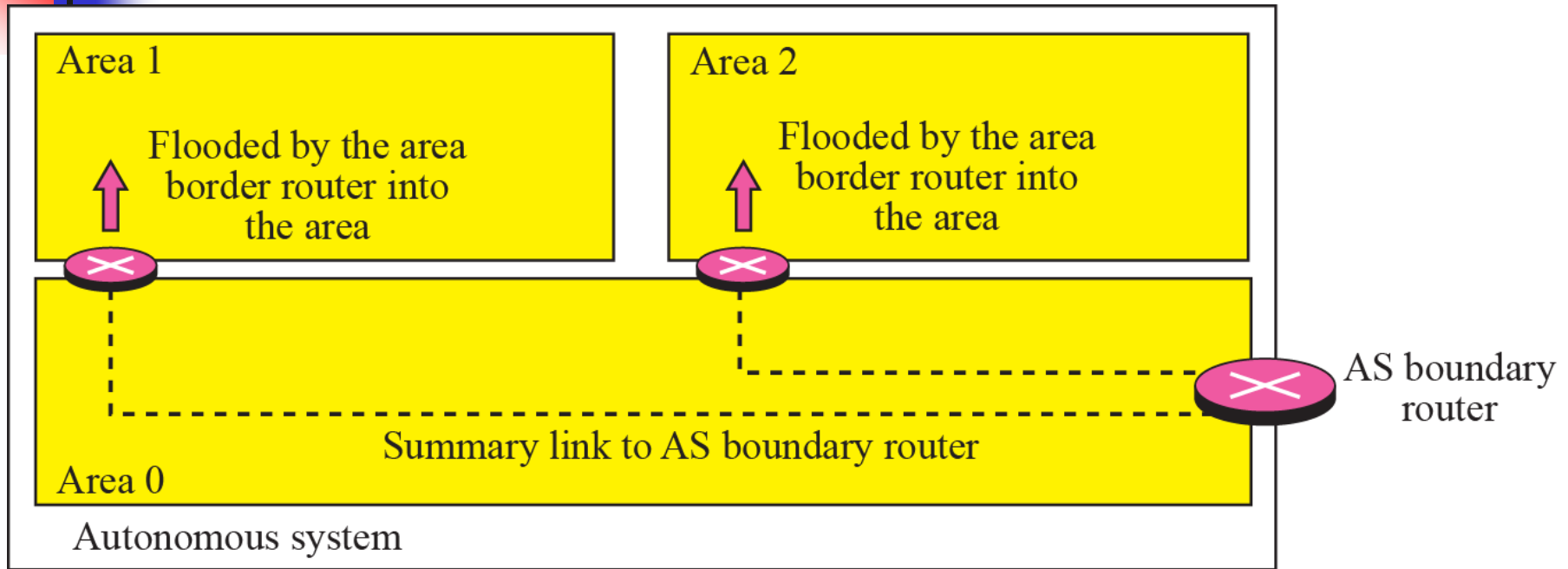
All three networks must advertise network links:

- Advertisement for N1 is done by R1 because it is the only attached router and therefore the designated router.
- Advertisement for N2 can be done by either R1, R2, or R3, depending on which one is chosen as the designated router.
- Advertisement for N3 is done by R3 because it is the only attached router and therefore the designated router

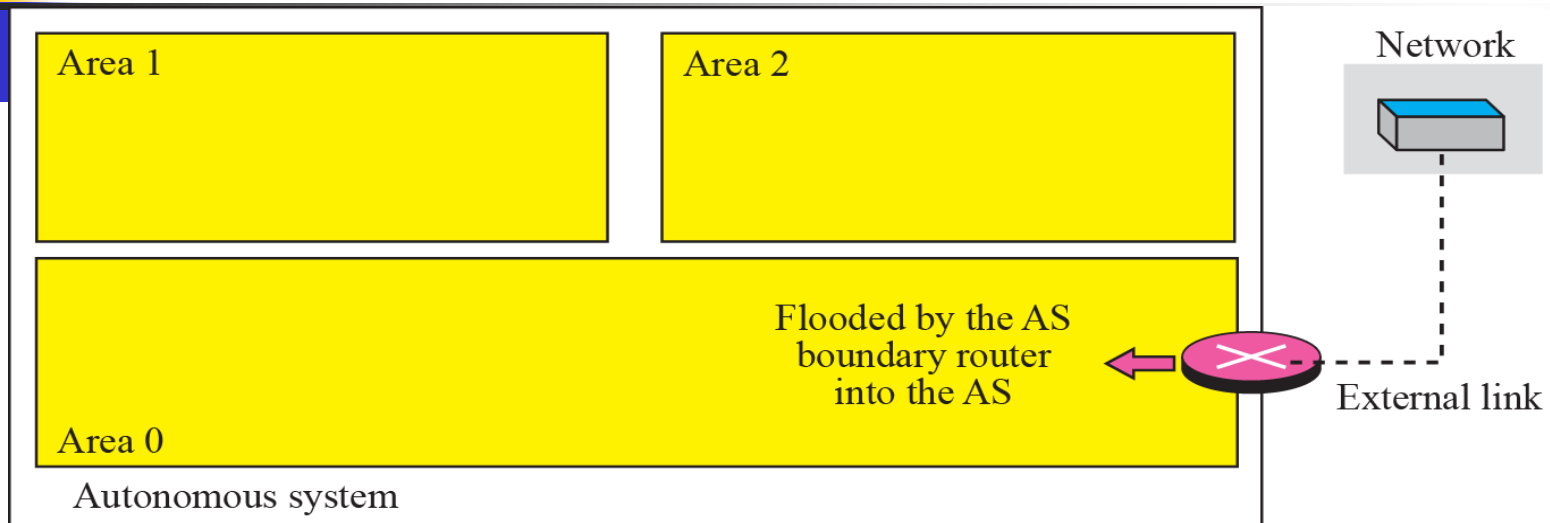
Summary link to network



Summary link to AS boundary router

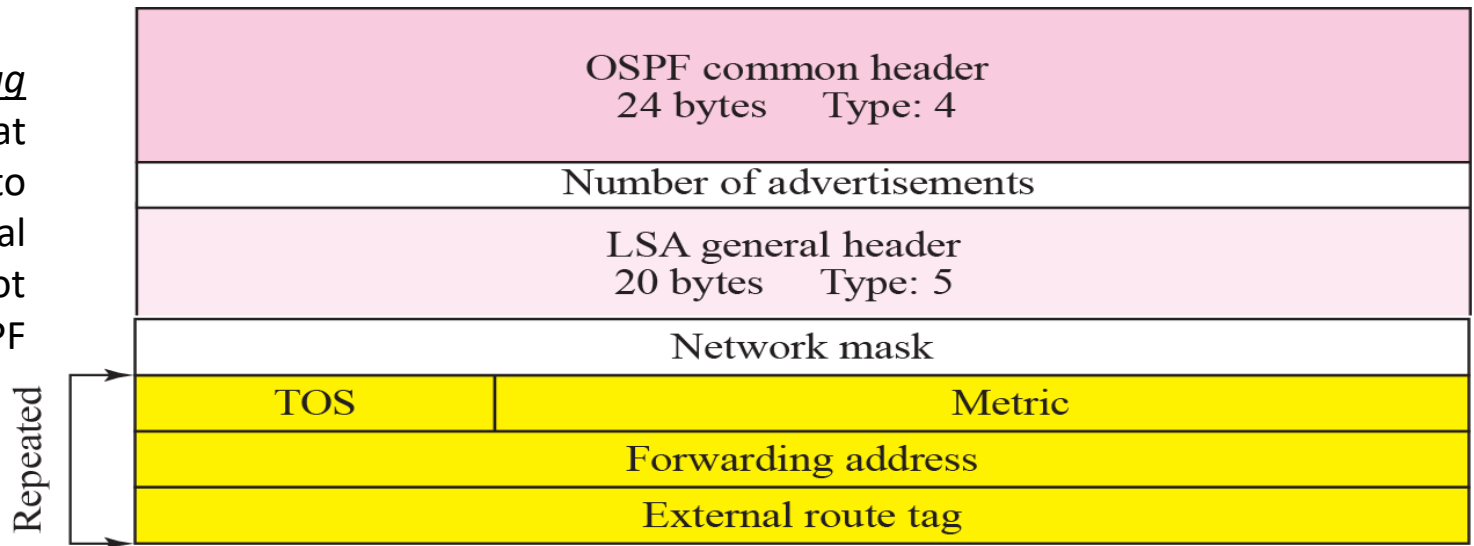


External link



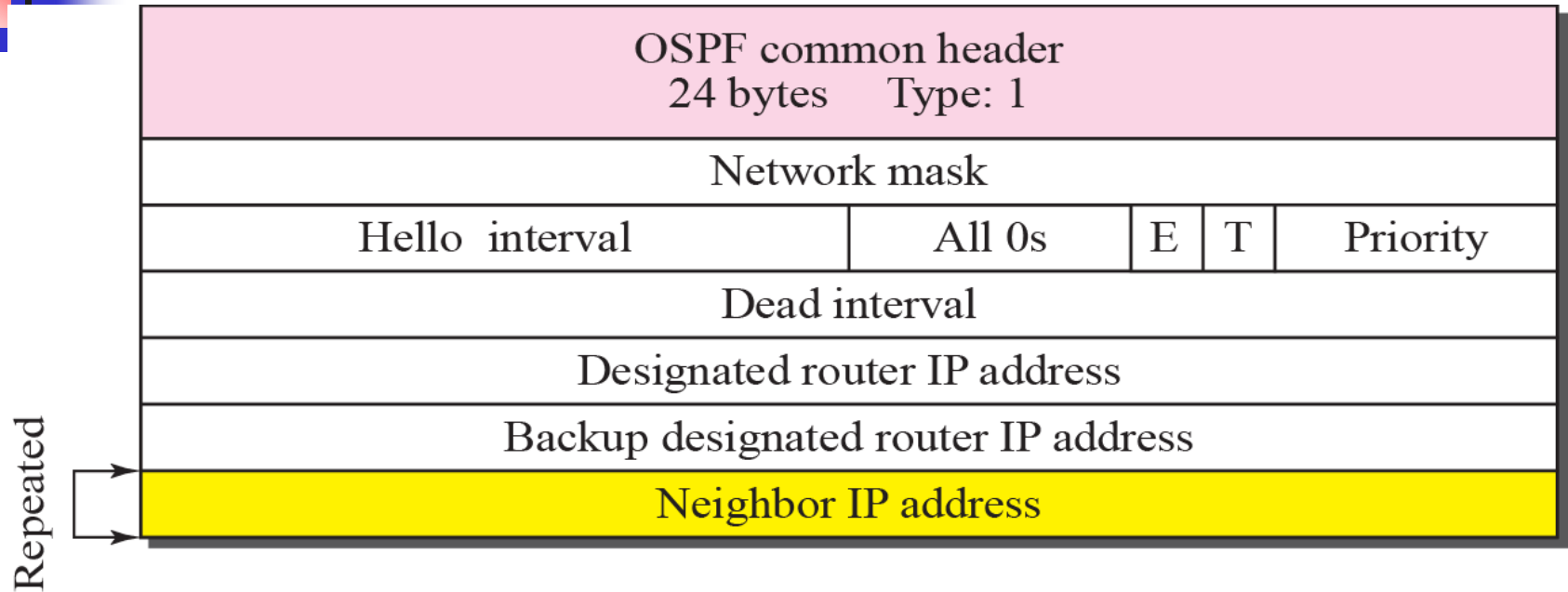
External Route Tag

is a 32-bit field that can be used to manage external routes, but is not used by the OSPF protocol itself



The primary purpose of the External Route Tag is to provide a means for identifying and managing external routes learned through redistribution into OSPF from other routing protocols.

Hello packet



OSPF uses the hello message to create neighborhood relationship and to test the reachability of neighbors.

This is the first step in link state routing. Before a router can flood all of the other routers with information about its neighbors, it must first greet its neighbors.

Hello packet

- ❑ **Network mask.** This 32-bit field defines the network mask of the network over which the hello message is sent.
- ❑ **Hello interval.** This 16-bit field defines the number of seconds between hello messages.
- ❑ **E flag.** This is a 1-bit flag. When it is set, it means that the area is a stub area.
- ❑ **T flag.** This is a 1-bit flag. When it is set, it means that the router supports multiple metrics.
- ❑ **Priority.** This field defines the priority of the router. The priority determines the selection of the designated router. After all neighbors declare their priorities, the router with the highest priority is chosen as the designated router. The one with the second highest priority is chosen as the backup designated router. If the value of this field is 0, it means that the router never wants to be a designated or a backup designated router.
- ❑ **Dead interval.** This 32-bit field defines the number of seconds that must pass before a router assumes that a neighbor is dead.
- ❑ **Designated router IP address.** This 32-bit field is the IP address of the designated router for the network over which the message is sent.
- ❑ **Backup designated router IP address.** This 32-bit field is the IP address of the backup designated router for the network over which the message is sent.
- ❑ **Neighbor IP address.** This is a repeated 32-bit field that defines the routers that have agreed to be the neighbors of the sending router. In other words, it is a current list of all the neighbors from which the sending router has received the hello message.

- **Hello Interval:**

This determines how often a router sends Hello packets to maintain neighbor relationships.

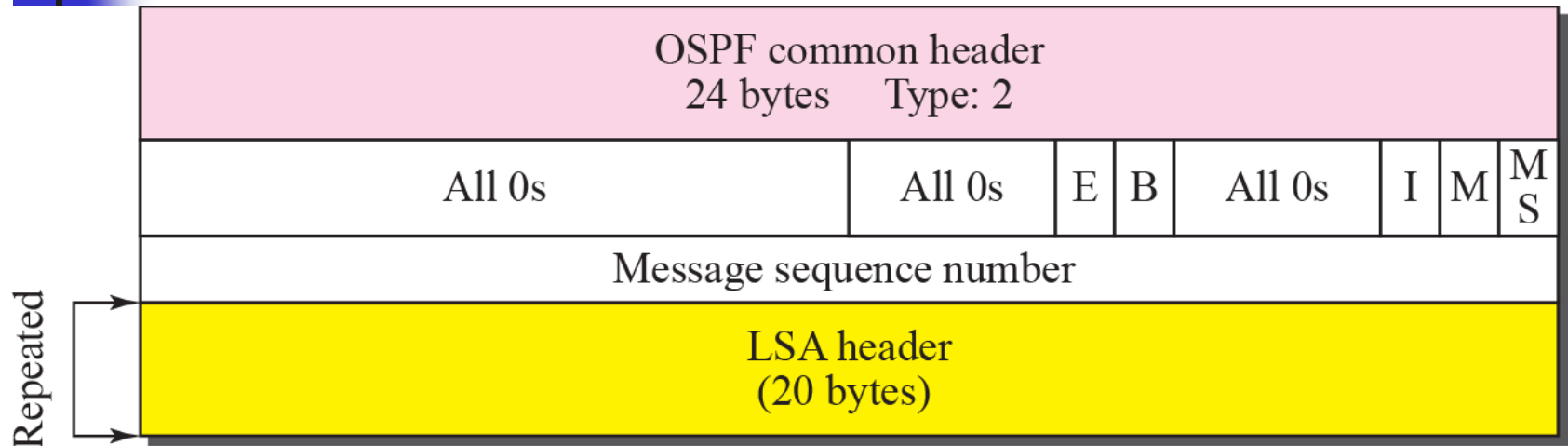
- **Dead Interval:**

This is the time a router waits to receive Hello packets from a neighbor before declaring that neighbor as down.

In OSPF,
default hello interval is 10 seconds to 30 seconds

default dead interval is four times the hello interval, so it's 40 seconds to 120 seconds

Database description packet



When a router is connected to the system for the first time or after a failure, it needs the complete link state database immediately.

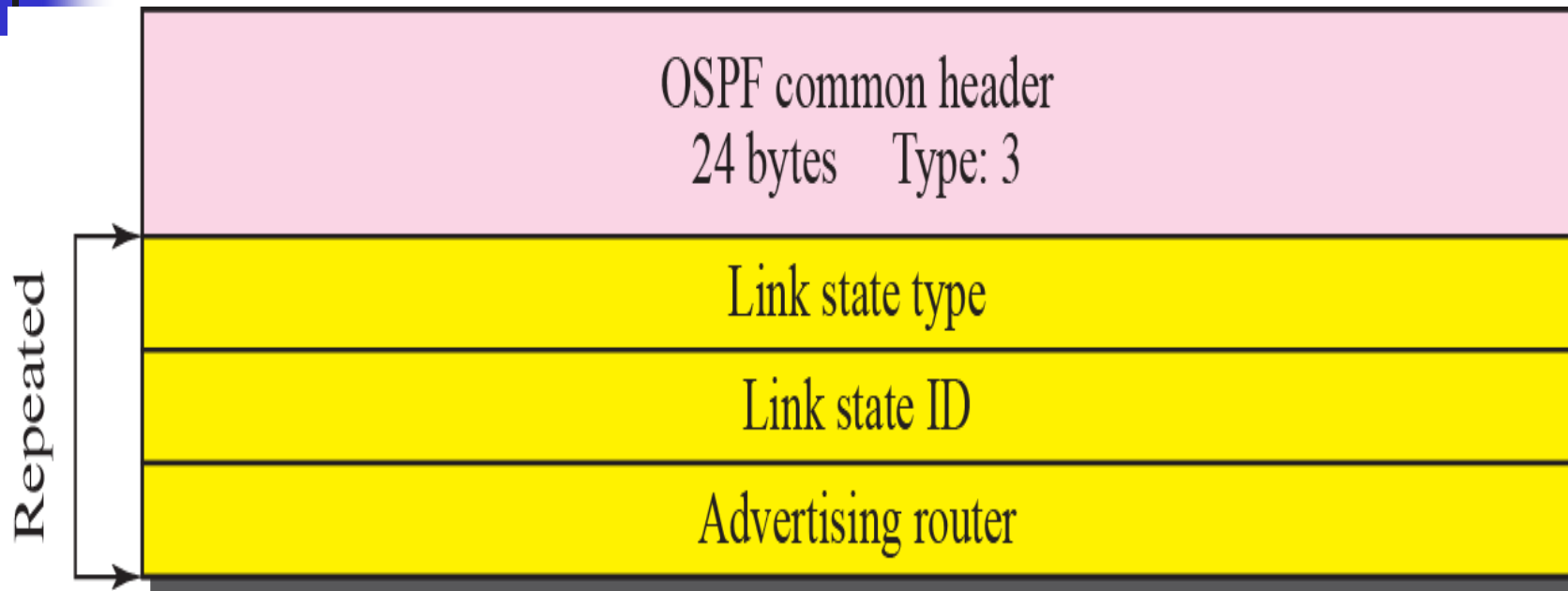
Therefore, it sends hello packets to greet its neighbors. If this is the first time that the neighbors hear from the router, they send a database description message.

The database description packet does not contain complete database information; it only gives an outline, the title of each lines in the database.

Database description packet

- ❑ **E flag.** This 1-bit flag is set to 1 if the advertising router is an autonomous boundary router (*E* stands for external).
- ❑ **B flag.** This 1-bit flag is set to 1 if the advertising router is an area border router.
- ❑ **I flag.** This 1-bit field, the *initialization* flag, is set to 1 if the message is the first message.
- ❑ **M flag.** This 1-bit field, the *more* flag, is set to 1 if this is not the last message.
- ❑ **M/S flag.** This 1-bit field, the *master/slave* bit, indicates the origin of the packet: master (M/S = 1) or slave (M/S = 0).
- ❑ **Message sequence number.** This 32-bit field contains the sequence number of the message. It is used to match a request with the response.
- ❑ **LSA header.** This 20-byte field is used in each LSA. The format of this header is discussed in the link state update message section. This header gives the outline of each link, without details. It is repeated for each link in the link state database.

Link state request packet



Link state age	Reserved	E	T	Link state type
Link state ID				
Advertising router				
Link state sequence number				
Link state checksum		Length		

LSA General Header

Link state type	Link state ID
Router link	IP address of the router
Network link	IP address of the designated router
Summary link to network	Address of the network
Summary link to AS boundary	IP address of the boundary router
External link	Address of the external network

Link state acknowledgment packet

OSPF common header
24 bytes Type: 5

LSA general header
20 bytes Corresponding type

OSPF packets are encapsulated in IP datagrams.

DVR	LSR
1980	1990
Bandwidth requirement is less	High bandwidth requirement
Local Knowledge	Global Knowledge
Bellman Ford / dynamic programming	Dijkstra Algorithm / greedy approach
Slow Convergence	Fast Convergence
Count to infinity	No count to infinity
Persistent loops	Transient loops
RIP	OSPF
Update only to neighbors	Use flooding for update

Interdomain Routing

Path vector Routing

Path vector routing

It is exterior routing protocol also known as interdomain or inter-AS routing.

- In distance vector routing, a router has a list of networks that can be reached in the same AS with the corresponding cost (number of hops).
- In path vector routing, a router has a list of networks that can be reached with the path (list of ASs to pass) to reach each one.
- domain of operation of the distance vector routing is a single AS.
- domain of operation of the path vector routing is the whole Internet.
- The distance vector routing tells us the distance to each network;
- the path vector routing tells us the path.

Analogy

A national map and an international map.

- ✓ A national map can tell us the **road to each city and the distance to be travelled** if we choose a particular route.
- ✓ an international map can tell us **which cities exist in each country and which countries should be passed** before reaching that city.

Reachability

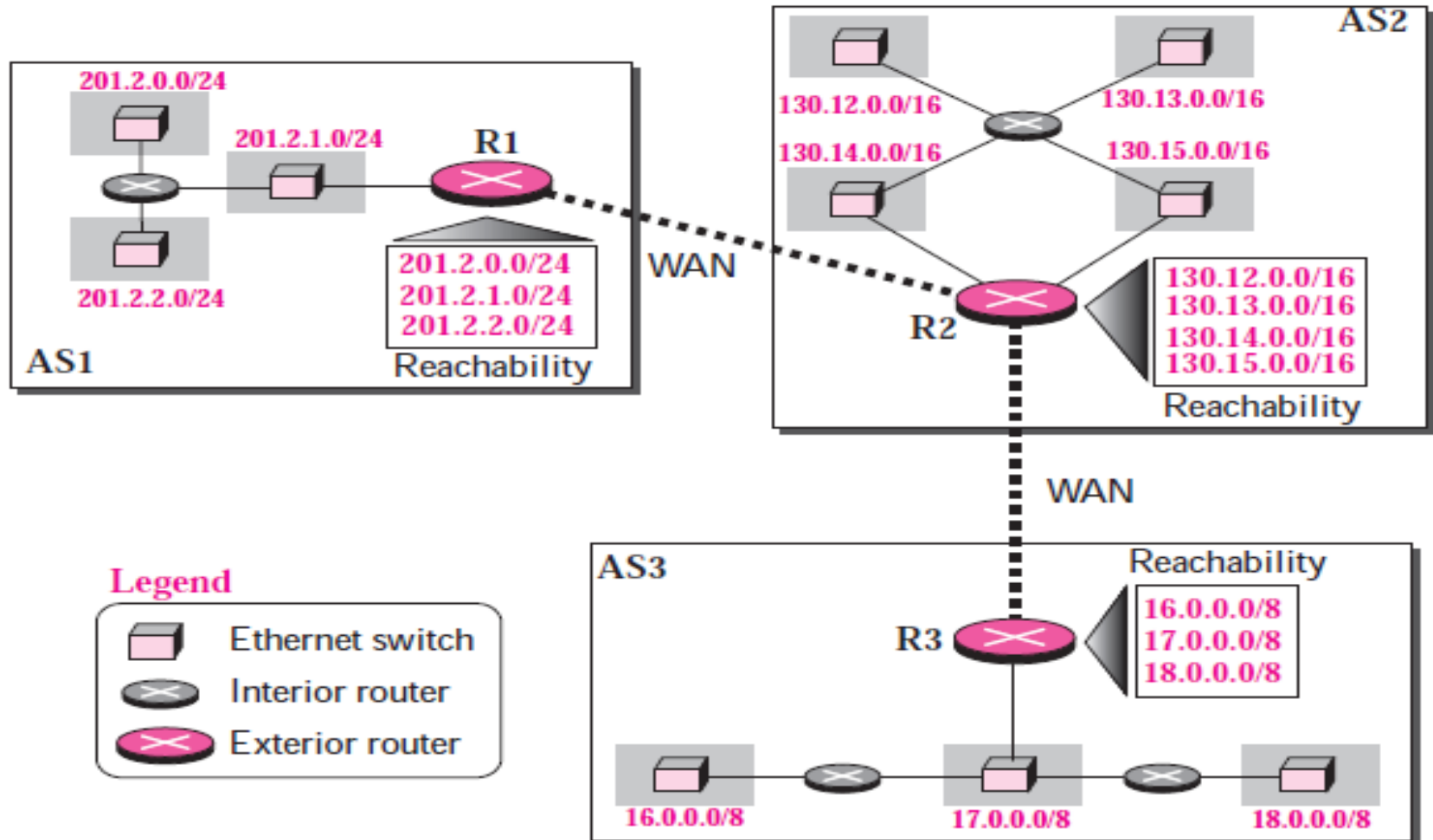
To be able to provide information to other ASs, each AS must have at least one path vector routing that collects ***reachability*** information about each network in that AS.

The information collected in this case only means which network, identified by its network address (CIDR prefix), exists (can be reached in this AS).

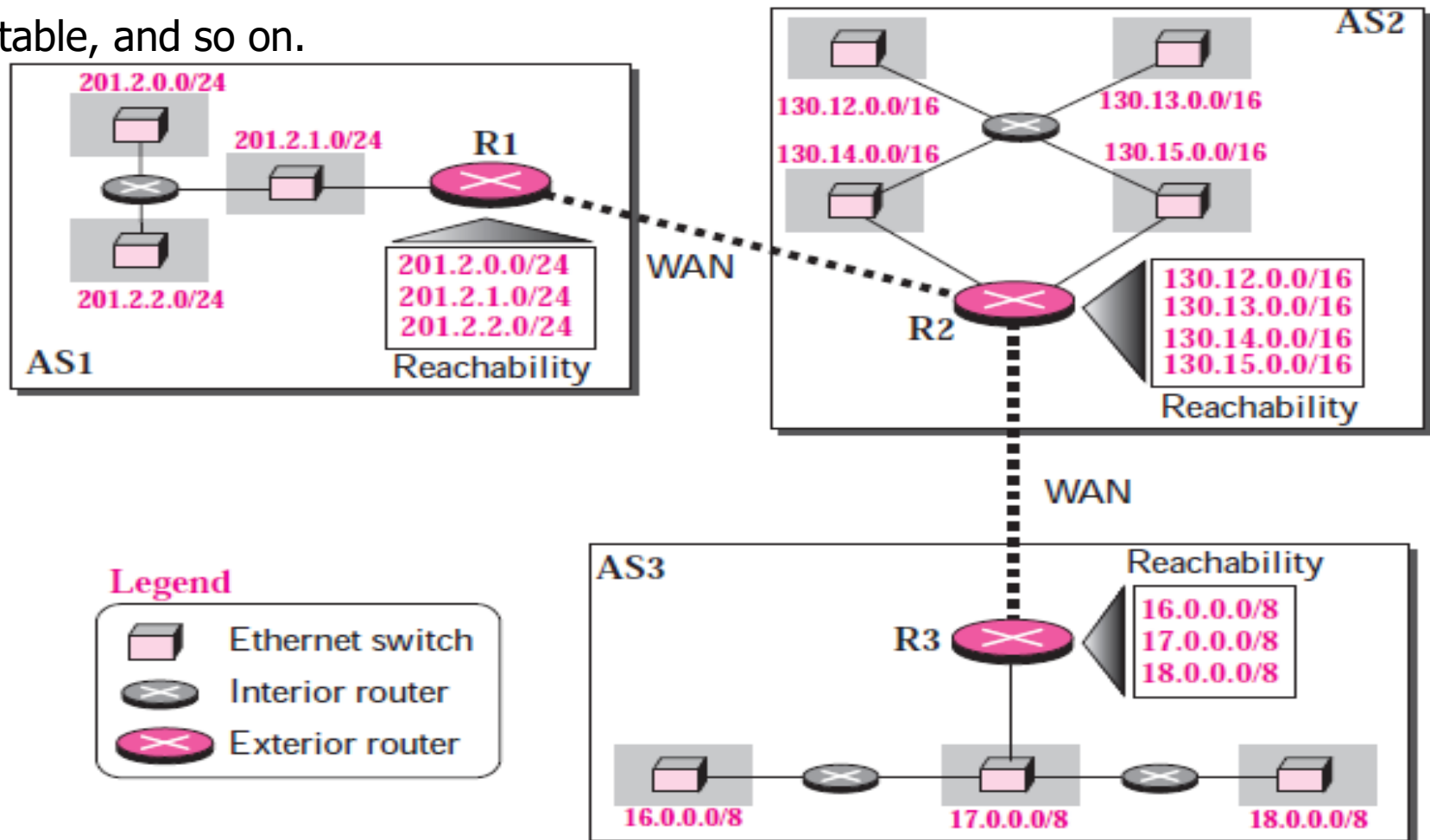
Reachability

Each AS needs to have a list of existing networks in its territory.

Figure shows three ASs. Each distance vector (exterior router) has created a list which shows which network is reachable in that AS.



- Router R1 in AS1 can send its reachability list to router R2.
- Router R2, after combining its reachability list, can send the result to both R1 and R3.
- Router R3 can send its reachability list to R2, which in turn improves its routing table, and so on.



Routing Tables

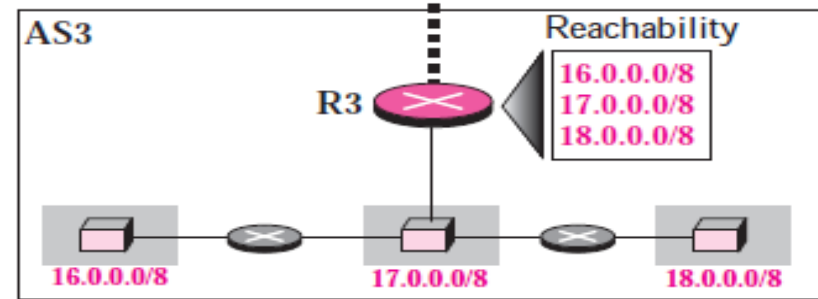
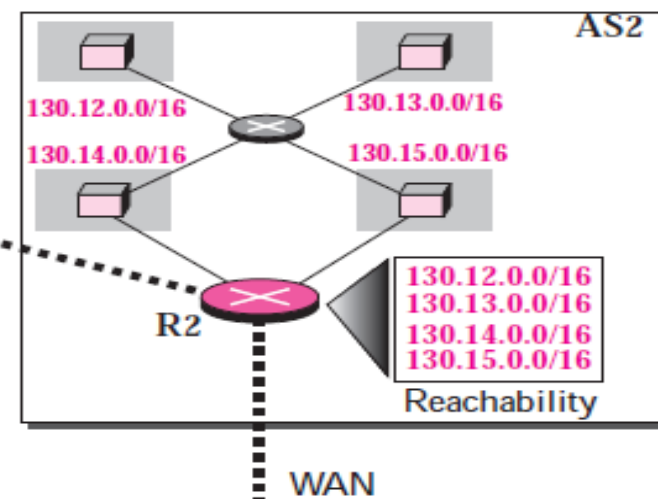
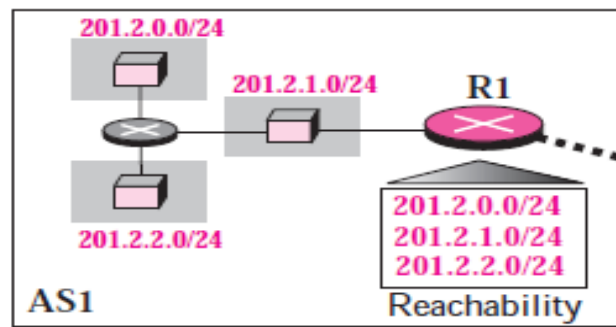
A path vector routing table for each router can be created if ASs share their reachability list with each other.

Ex.

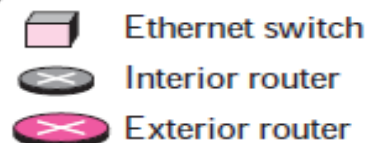
- Router R1 in AS1 can send its reachability list to router R2.
- Router R2, after combining its reachability list, can send the result to both R1 and R3.
- Router R3 can send its reachability list to R2, which in turn improves its routing table, and so on.

Routing table for each router after all three routers have updated their routing table.

R1		R2		R3	
					
Network	Path	Network	Path	Network	Path
201.2.0.0/24	AS1 (This AS)	201.2.0.0/24	AS2, AS1	201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS1 (This AS)	201.2.1.0/24	AS2, AS1	201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS1 (This AS)	201.2.2.0/24	AS2, AS1	201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS1, AS2	130.12.0.0/16	AS2 (This AS)	130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS1, AS2	130.13.0.0/16	AS2 (This AS)	130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS1, AS2	130.14.0.0/16	AS2 (This AS)	130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS1, AS2	130.15.0.0/16	AS2 (This AS)	130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS1, AS2, AS3	16.0.0.0/8	AS2, AS3	16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS1, AS2, AS3	17.0.0.0/8	AS2, AS3	17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS1, AS2, AS3	18.0.0.0/8	AS2, AS3	18.0.0.0/8	AS3 (This AS)
Path-Vector Routing Table		Path-Vector Routing Table		Path-Vector Routing Table	



Legend



R1

Network	Path
201.2.0.0/24	AS1 (This AS)
201.2.1.0/24	AS1 (This AS)
201.2.2.0/24	AS1 (This AS)
130.12.0.0/16	AS1, AS2
130.13.0.0/16	AS1, AS2
130.14.0.0/16	AS1, AS2
130.15.0.0/16	AS1, AS2
16.0.0.0/8	AS1, AS2, AS3
17.0.0.0/8	AS1, AS2, AS3
18.0.0.0/8	AS1, AS2, AS3

Path-Vector Routing Table

R2

Network	Path
201.2.0.0/24	AS2, AS1
201.2.1.0/24	AS2, AS1
201.2.2.0/24	AS2, AS1
130.12.0.0/16	AS2 (This AS)
130.13.0.0/16	AS2 (This AS)
130.14.0.0/16	AS2 (This AS)
130.15.0.0/16	AS2 (This AS)
16.0.0.0/8	AS2, AS3
17.0.0.0/8	AS2, AS3
18.0.0.0/8	AS2, AS3

Path-Vector Routing Table

R3

Network	Path
201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS3 (This AS)

Path-Vector Routing Table

R1



Network	Path
201.2.0.0/24	AS1 (This AS)
201.2.1.0/24	AS1 (This AS)
201.2.2.0/24	AS1 (This AS)
130.12.0.0/16	AS1, AS2
130.13.0.0/16	AS1, AS2
130.14.0.0/16	AS1, AS2
130.15.0.0/16	AS1, AS2
16.0.0.0/8	AS1, AS2, AS3
17.0.0.0/8	AS1, AS2, AS3
18.0.0.0/8	AS1, AS2, AS3

Path-Vector Routing Table

R2



Network	Path
201.2.0.0/24	AS2, AS1
201.2.1.0/24	AS2, AS1
201.2.2.0/24	AS2, AS1
130.12.0.0/16	AS2 (This AS)
130.13.0.0/16	AS2 (This AS)
130.14.0.0/16	AS2 (This AS)
130.15.0.0/16	AS2 (This AS)
16.0.0.0/8	AS2, AS3
17.0.0.0/8	AS2, AS3
18.0.0.0/8	AS2, AS3

Path-Vector Routing Table

R3



Network	Path
201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS3 (This AS)

Path-Vector Routing Table

How the Router R1 will handle if a packet arrives for the network **201.2.2.0/24** ?

How the Router R1 will handle if a packet arrives for the network **130.14.0.0/16** ?

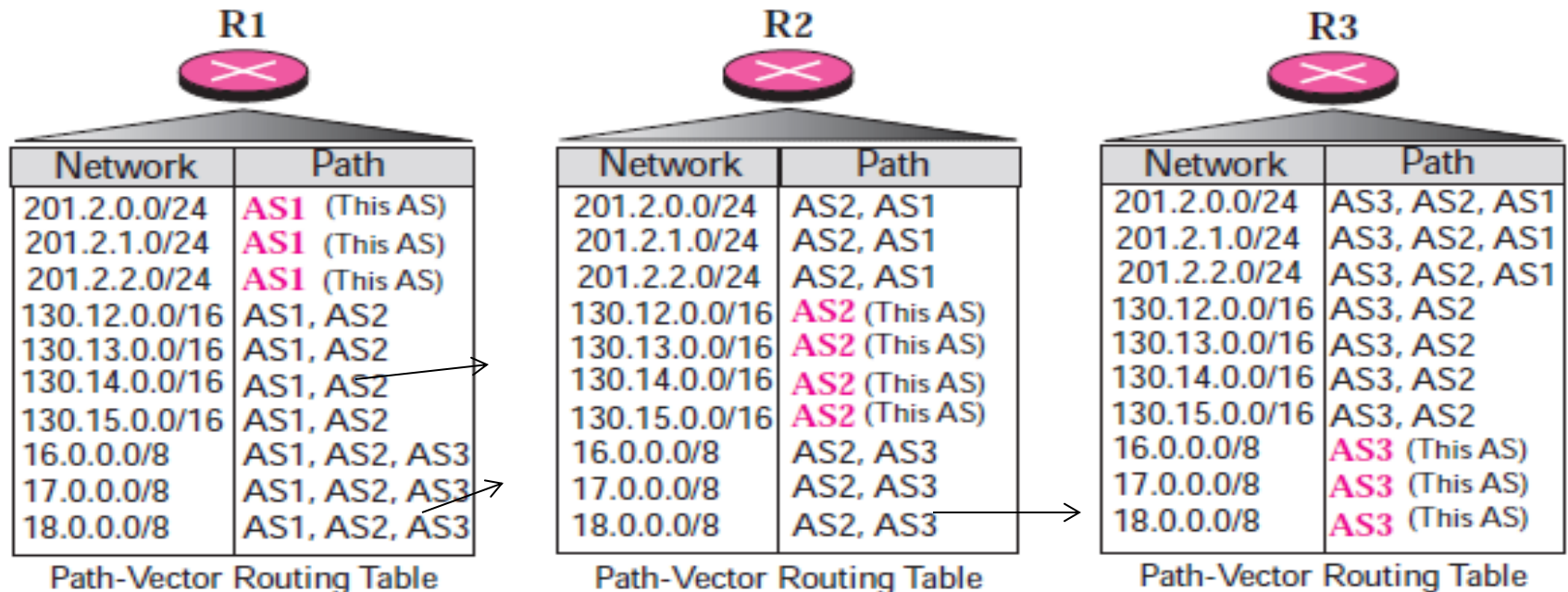
How the Router R3 will handle if a packet arrives for the network **201.2.1.0/24** ?

Loop Prevention

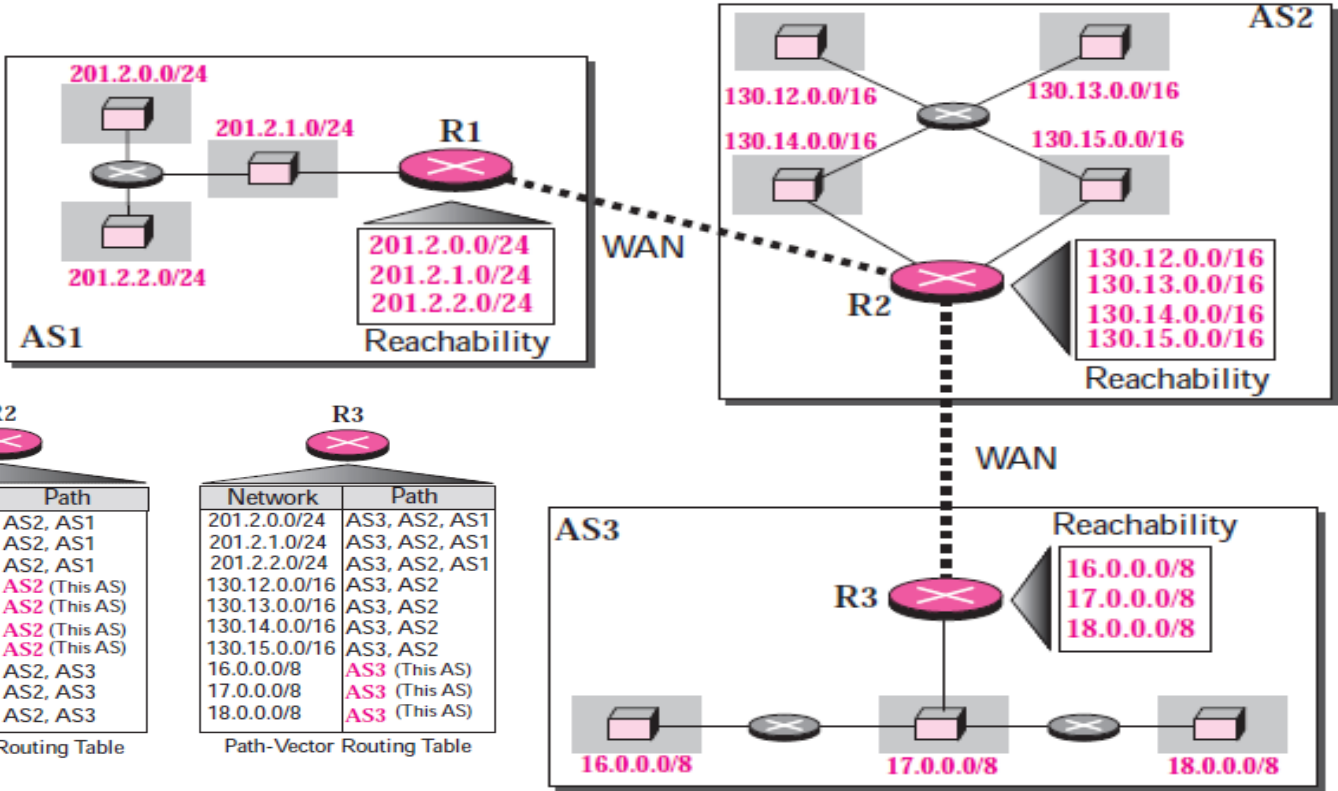
The instability of distance vector routing and the creation of loops can be avoided in path vector routing.

When a router receives a reachability information -

- ✓ It checks to see if its autonomous system is in the path list to any destination.
- ✓ If it is, looping is involved and that network-path pair is discarded.



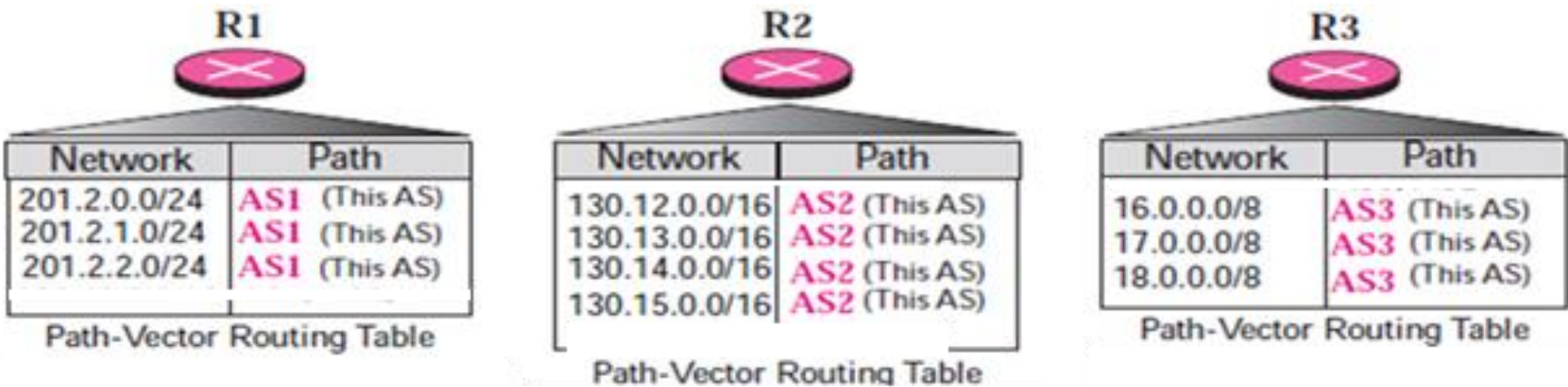
Loop Prevention



Whats happens if R1 send R2

130.14.0.0/16 | AS1, AS2

Initial Path Vector Routing Table



Aggregation

The path vector routing protocols normally support CIDR notation and the aggregation of addresses (if possible).

This helps to make the path vector routing table simpler and exchange between routers faster.

R1		R2		R3	
Path-Vector Routing Table		Path-Vector Routing Table		Path-Vector Routing Table	
Network	Path	Network	Path	Network	Path
201.2.0.0/24	AS1 (This AS)	201.2.0.0/24	AS2, AS1	201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS1 (This AS)	201.2.1.0/24	AS2, AS1	201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS1 (This AS)	201.2.2.0/24	AS2, AS1	201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS1, AS2	130.12.0.0/16	AS2 (This AS)	130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS1, AS2	130.13.0.0/16	AS2 (This AS)	130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS1, AS2	130.14.0.0/16	AS2 (This AS)	130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS1, AS2	130.15.0.0/16	AS2 (This AS)	130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS1, AS2, AS3	16.0.0.0/8	AS2, AS3	16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS1, AS2, AS3	17.0.0.0/8	AS2, AS3	17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS1, AS2, AS3	18.0.0.0/8	AS2, AS3	18.0.0.0/8	AS3 (This AS)

Routing table after aggregation

R1		R2		R3	
Path-Vector Routing Table		Path-Vector Routing Table		Path-Vector Routing Table	
Network	Path	Network	Path	Network	Path
201.2.0.0/22	AS1 (This AS)	201.2.0.0/22	AS2, AS1	201.2.0.0/22	AS3, AS2, AS1
130.12.0.0/16	AS1, AS2	130.12.0.0/16	AS2 (This AS)	130.12.0.0/16	AS3, AS2
16.0.0.0/6	AS1, AS2, AS3	16.0.0.0/6	AS2, AS3	16.0.0.0/6	AS3 (This AS)

Policy Routing

Policy routing can be implemented through path vector routing.

When a router receives a message, it can check the path. If one of the autonomous systems listed in the path is against its policy, it can ignore that path and that destination.

- ❖ It does not update its routing table with this path, and
- ❖ it does not send this message to its neighbors.

Lecture videos on the topic can be found at

Link State Routing (LSR) and Open Shortest Path First (OSPF) - Update Packet Format

<https://www.youtube.com/watch?v=TDYg-bM8nEU>

Open Shortest Path First (OSPF) - Different Packet Format [Part -2]

<https://www.youtube.com/watch?v=EwkiisRC-yY>